

YAMAHA SUPPORT SOFTWARE

RCX-Studio 2020

User's Manual

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Introduction

About this manual

This manual describes how to install RCX-Studio 2020 support software for YAMAHA RCX series robot controllers (simply called "RCX-Studio 2020" or "this software" from now on) as well as major functions you can run with this software. Before installing RCX-Studio 2020 be sure to read this manual and follow the instructions.

Software license agreement

This software may be used only when you agree to the terms stipulated below.

By using this software or installing it in a personal computer (PC), you are consenting to the terms stipulated below.

Please read the following terms carefully.

1. The copyrights to this software are owned by YAMAHA MOTOR CO., LTD., IM Operations (hereafter "our company").
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4. Reverse engineering, decompiling, altering or modifying this software is prohibited.
5. Transfer, rental or leasing of this software to a third party, whether for profit or not, is prohibited without permission from our company.
6. Our company takes no responsibility for any consequences arising from the use of this software or this manual.
7. Although our company has taken every reasonable effort with this software and this manual, we cannot guarantee that they are completely free from errors.
8. The specifications of this software and the content of this manual are subject to change without notice in the future.

CAUTION

For information on how to operate the robot controller and the robot, carefully read the manuals for the controller and robot that you are using, and follow the instructions given in those manuals.

RCX-Studio 2020 package contents

• License agreement ... 1 pce.

A document proving that you have received a software license. Be sure to store this document with great care.

• USB key ... 1 pce.

A blue USB key is supplied with the Basic package while a purple USB key is supplied with the Pro package.

Available functions may vary depending on the USB key you are using.

For details about the USB key, refer to "1. Overview of RCX-Studio 2020".

System requirements

The following hardware and software are required to install and use RCX-Studio 2020.

Operating environment	Operating system (OS)	Microsoft Windows 7 SP1 (32/64bit) / 8.1 (32/64bit) / 10 (32/64bit) / 11
	Execution environment	.NET Framework 4.5 or more
	CPU	Recommended: Intel Core i5 2 GHz or more, Minimum: Intel Celeron 2 GHz or more, 3D-SIM is invalid.: Intel Core 2 Duo 2 GHz or more
	RAM memory	Recommended: 8 GB or more, Minimum: 4 GB or more, 3D-SIM is invalid: 1 GB or more
	Hard disk storage space	1GB of available space required on installation drive
	Communication port	Communication cable: Serial communication port, Ethernet port, or USB port
	Others	Dedicated commutation cable (For D-Sub or USB) Ethernet cable (category 5 or better) USB port: 1 port (For USB key)
Applicable controller		RCX340 / RCX341 / RCX320 (*)
Applicable robot		YAMAHA robot that can be connected to the RCX340, RCX320.

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* On RCX-Studio, "RCX340" is displayed even when RCX341 is used, but there is no functional problem.

Safety Precautions

Before using this product, be sure to read this manual carefully as well as the robot controller user's manual and programming manual. Take sufficient precautions to ensure safety and handle the product correctly.

The cautions given in this manual are related to this product. Refer to the robot controller user's manual for details on the cautions to be taken with the robot controller system using this product.

The safety precautions are ranked as "WARNING" and "CAUTION" in this manual.



WARNING

This indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.



CAUTION

This indicates a potentially hazardous situation which, if not avoided, could result in minor or moderate injury, or damage to the equipment.

Note that some items described as "CAUTION" may lead to serious results depending on the situation. In any case, important information that must be observed is explained.

Store this manual where it can be easily referred to, and make sure that it is delivered to the end user.

Refer to the manual by any of the following methods when installing, operating or adjusting the robot and controller.

1. Install, operate or adjust the robot and controller while viewing the manual on your computer screen.
2. Install, operate or adjust the robot and controller while referring to a printout of the necessary pages from the manual.
3. Install, operate or adjust the robot and controller while referring to the printed version of the manual (available for an additional fee).

Manuals are available by downloading from our website (Member Site). Registration is required to access the Member Site:

https://www2.yamaha-motor.co.jp/Robot/Member/loginagain/lang_div/en

1. Overview of RCX-Studio 2020

RCX-Studio 2020 is operation support software for RCX3-series robot controllers.

In addition to the functionality of previous software, it provides a user interface that is easier to use as well as being more powerful.

In order to take full advantage of RCX-Studio 2020 functionality, be sure to read the contents of this manual.



WARNING

When operating a robot using RCX-Studio 2020, make sure that all workers are outside the safety fence, and then the emergency stop is ready to input at any time.

Compatible robot controllers

RCX-Studio 2020 supports the following YAMAHA multi-axis robot controllers.

- RCX340
- RCX341
- RCX320

USB key

A USB key is supplied to the RCX-Studio 2020 to prevent robot operation mistakes.

The functions are restricted as described in the list shown below when the USB key is not connected to the PC or in accordance with the type of the USB key.

Functions		When the USB key is not connected.	Basic (blue) ^{*2}	Pro (purple) ^{*2}
Backup/Restore by data transfer with controller		Valid ^{*1}	Valid	Valid
Controller operation in online mode		Invalid	Valid	Valid
File read (opening data)		Invalid	Valid	Valid
File save		Invalid	Valid	Valid
Real Time Trace		Only data save is invalid.	Valid	Valid
Cycletime Calculator		Invalid	Valid	Valid
iVY2 editor		Invalid	Valid	Valid
Data Difference		Except data saving	Valid	Valid
3D simulator function		Only capturing is invalid.	Valid	Valid
Custom window		Valid	Valid	Valid
Program template		Only file output is invalid.	Valid	Valid
CAD data read	STL, OBJ, VRML	Valid	Valid	Valid
	STEP	Invalid	Invalid	Valid
CAD to point conversion		Invalid	Invalid	Valid

^{*1}) USB key connection is required for backup when connected in offline mode. ^{*2}) USB key color



NOTE

- For details about the 3D simulator function, refer to the Help screen provided on the RCX-Studio 2020 functions.
- For details about the online and inline modes, refer to "4. Controller operation".

2. Installing/uninstalling RCX-Studio 2020

In order to use RCX-Studio 2020, you must install RCX-Studio 2020 on the PC. If an earlier version of RCX-Studio 2020 is already installed, you can upgrade to the new version without uninstalling the previous version.

The latest installer for RCX-Studio 2020 can be obtained from the member site in Yamaha Robot Website:
<https://global.yamaha-motor.com/business/robot>



NOTE

It is required to register as a member to download RCX-Studio 2020 from the member site.

2.1 To install RCX-Studio 2020

The following Steps describe how to install RCX-Studio 2020 in a PC running on Windows 7.

Step 1 Exit all the currently-running programs on the PC.

Step 2 Start the setup program.

Double-click (RCX-Studio 2020) you have downloaded. The setup program starts automatically. Depending on the PC settings, the icon name might be (RCX-Studio 2020.exe).

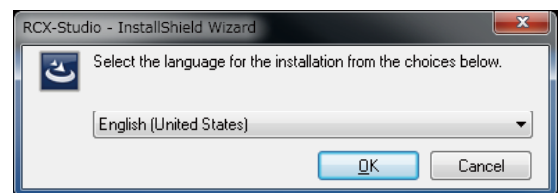


NOTE

There is only one type of RCX-Studio 2020 installer. Regardless of the type of the USB key ("purple" for Pro and "blue" for Basic), use the same installer.

Step 3 Install RCX-Studio 2020.

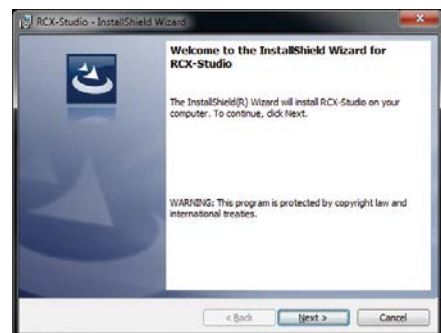
The installer wizard shows a screen that lets you choose the display language. Choose the language to display on the wizard and then click (OK). Either (Japanese) or (English) can be chosen as the language.



NOTE

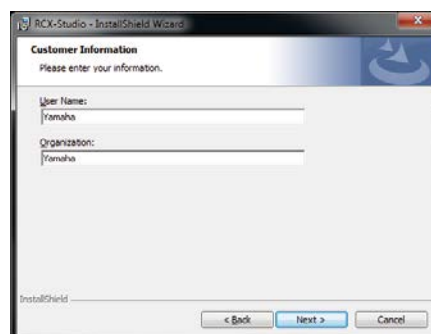
- If you choose (English), all subsequent screens of the wizard are displayed in English.
- The language setting is also applied to updates and to the uninstall wizard screens.

Step 4 Click [Next] at the [Welcome to the InstallShield Wizard for RCX-Studio 2020] screen.



Step 5 The [Customer Information] screen.

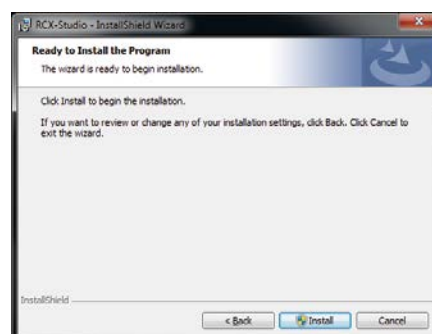
Enter your user name and organization, and click (Next).

**Step 6** At the [Destination Folder] screen, specify the folder in which you want to install RCX-Studio 2020.

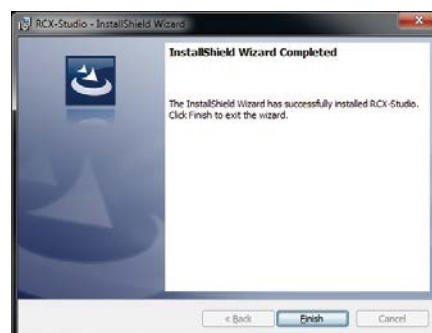
By default, (C:\Program Files\Yamaha Motor\RCX-Studio 2020\) is selected. To change the installation destination, click (Change) and specify the installation destination. When you have finished specifying the installation destination, click (Next).

**Step 7** At the [Language selection for RCX-Studio 2020] screen, choose the language that you want RCX-Studio 2020 to display, and click [Next].

Language options are (Japanese), (English), and (Chinese). The language displayed by RCX-Studio 2020 can be changed even after installation.

**Step 8** Installing begins by clicking [Install] on the screen.**Step 9** On the screen appears when installation is completed, click [Finish] to exit the wizard.**TIP**

Depending on the PC settings, restarting is necessary.



2.2 To upgrade RCX-Studio 2020

If an earlier version of RCX-Studio 2020 has been installed on the PC, you can upgrade to the new version. The following Steps describe how to upgrade in a PC running on Windows 7.

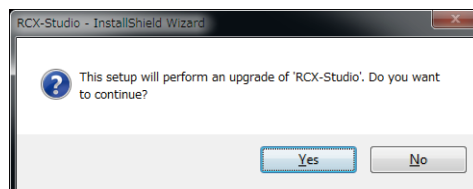
Step 1 Exit all currently-running programs.

Step 2 Start the setup program.

Double-click (RCX-Studio 2020) you have downloaded. The setup program starts automatically. Depending on the PC settings, the icon name might be (RCX-Studio 2020.exe).

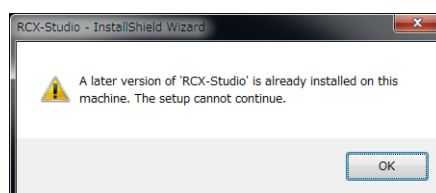
Step 3 Upgrade RCX-Studio 2020.

A message asks you whether to upgrade. To execute the upgrade, click (Yes). If not, click (No) to exit the upgrade.

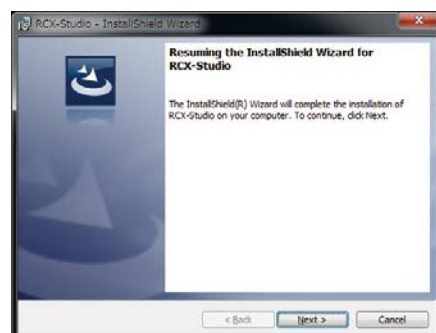


NOTE

- If the version of RCX-Studio 2020 installed on the PC is later than that on the setup program, you can't perform the upgrade. In this case, uninstall RCX-Studio 2020 from the PC and then install it.
- If the version of RCX-Studio 2020 installed on the PC is the same as that on the setup program, the program maintenance screen appears. In the screen, you can modify, repair, or remove RCX-Studio 2020.

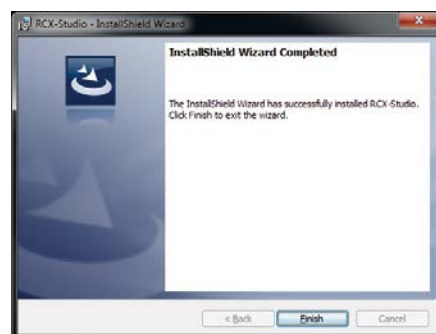


Step 4 At the screen shown at right, click [Next] to start the RCX-Studio 2020 upgrade.



Step 5 When the upgrade is completed, the screen shown at right appears.

Click (Finish) to exit the wizard.



2.3 To uninstall RCX-Studio 2020

You can uninstall RCX-Studio 2020 from your PC in either of the following two ways.
The following Steps describe how to uninstall in a PC running on Windows 7.

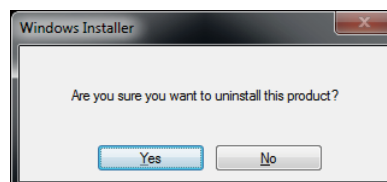
■ Using [Uninstall RCX-Studio 2020]

Step 1 Select [Uninstall RCX-Studio 2020].

From the (Start) menu, click (All Programs)-(RCX-Studio 2020)-(Uninstall RCX-Studio 2020).

Step 2 Uninstall RCX-Studio 2020.

At the uninstall confirmation message, click (Yes).



■ Using [Programs and Features]

Step 1 Open the [Programs and Features] window.

From the (Start) menu, click (Control Panel) and then click (Uninstall a Program).

TIP

If you've selected (Large Icons) or (Small Icons) as the view in Control Panel, click (Programs and Features).

Step 2 Select the RCX-Studio 2020.

Step 3 Uninstall RCX-Studio 2020.

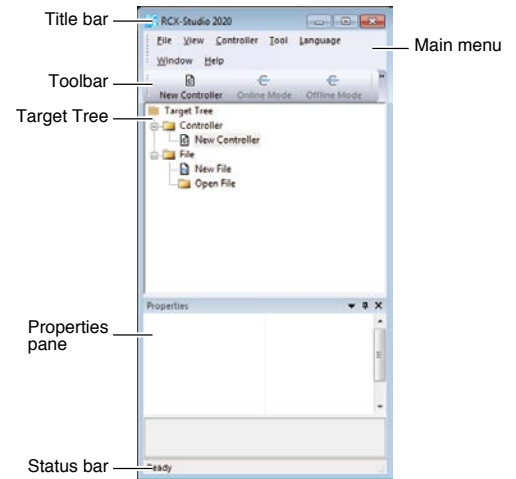
Follow the on-screen instructions to uninstall the RCX-Studio 2020.

3. RCX-Studio 2020 window

RCX-Studio 2020 has the following two types of window. The following section describes each type.

3.1 RCX-Studio 2020 window

When starting RCX-Studio 2020, the window shown at right appears. Connecting to the controller and administrating files are mainly performed on this window.



■ Description of the RCX-Studio 2020 window

Main menu

◆ File

New File	Creates a new file.
Open File	Opens a file.
Open Alarm History File	Opens an alarm history file.
Open Warning History File	Opens a warning history file.
Open Project	Opens a project.
Save Project As Name	Saves the added controller and file settings as a project with the name that you specify.
Exit	Exits RCX-Studio 2020.

* For details on files, refer to "5. File operation "

◆ View

Toolbar	Toolbar	Shows/hides the toolbar.
	Small Size Icons	Shows small icons on the toolbar.
	Large Size Icons	Shows large icons on the toolbar.
	Show Text Labels	Shows/hides text on the toolbar.
Properties	Shows/hides the property pane.	
Status Bar	Shows/hides the status bar.	



NOTE

When you change the (Icon Size) or (Show Text Labels) setting for the toolbar, the setting also applies to the toolbar of the target window described later.

◇ Controller

New Controller	Creates a new controller.
Online	Starts the controller selected in the target tree in the online mode (actual machine connection). This menu item is valid when a controller that is not started or a controller that is started in the offline mode is selected.
Offline	Starts the controller selected in the target tree in the offline mode (emulator connection). This menu item is valid when a controller that is not started or a controller that is started in the online mode is selected.

◇ Tool

Data Transfer	Opens the Data Transfer dialog box, allowing you to transfer data from the controller to the PC or vice versa.
Save Alarm History	Saves the alarm history file.
Save Warning History	Saves the warning history file.
Data difference	Opens the Data Difference dialog box, allowing you to compare data between the controllers and files.
Cycletime Calculator	Opens the Cycletime Calculator dialog box, allowing you to calculate the robot cycle time.
Program Converter	Converts a program for the RCX2xx controller into that for the RCX3xx controller.
Option	Opens the Options dialog box, allowing you to specify options.

◇ Language

English	Sets the display language to English.
日本語	Sets the display language to Japanese.
中国語	Sets the display language to Chinese.



NOTE
If you change the language setting, restart RCX-Studio 2020.

◇ Window

Tile Windows	Displays the currently-shown target windows without overlapping.
Cascade Windows	Displays the currently-shown target windows overlapping.

◇ Help

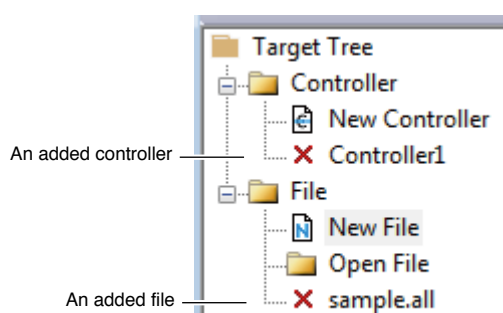
Help	Displays the Help window.
About RCX-Studio 2020	Shows the software version of RCX-Studio 2020.

Toolbar

The toolbar contains [New Controller], [Online], [Offline], [New File], and [Open File] buttons. These buttons execute the operations, "new controller", "start a controller in online mode", "start a controller in offline mode", "create a new file," and "open a file", respectively.

Target Tree

You can make a new connection to a controller, create a new file, or open a file from the target tree. The target tree shows controllers you added as new connections and files that you opened. By double-clicking a controller or file that is shown in the target tree, you can start the selected controller or open the file.



Properties

In Properties pane you can view detailed information about the controller or file that is selected in the target tree, and edit some of its settings.

*Properties cannot be edited while connecting to the controller selected on the target tree or opening the selected files.

Status bar

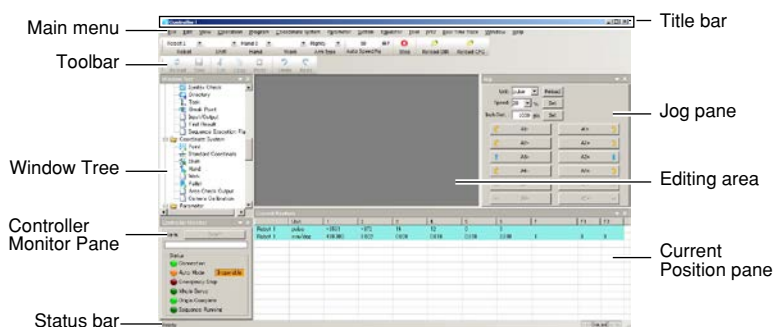
The status bar shows explanation of the functions that the mouse is pointing and the status of special keys.

3.2 Target window

The target window appears when you connect to a robot controller (controller operation), create a new file, or open an existing file (file operation). During controller operation, the robot status such as each axis jog movement, robot operation, or I/O monitoring can be referred to or controlled. During file operation, each data is edited.

Target window (controller operation)

Connect to the controller to display the target window.

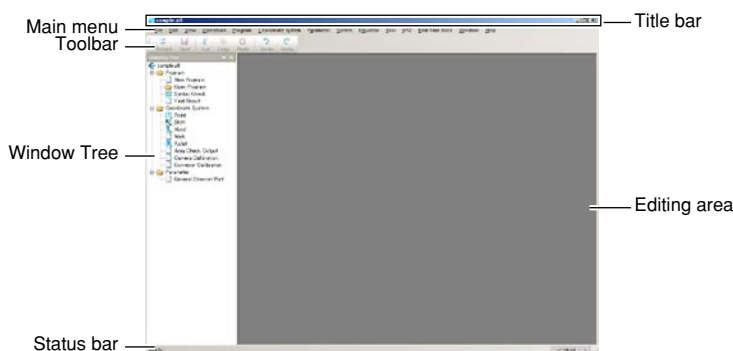


TIP

For details on how to connect to a controller, refer to "4.2 Adding a controller"

Target window (file operation)

Create a new file or open an existing file to display the target window.



TIP

For details on how to create a new file or open a file, refer to "5.1 Creating a new data file" and "5.2 Opening a data file".

■ Description of the target window

Title bar

The title bar shows the controller name that was specified at the time of connection.

Main menu

The target window (controller) provides the following menus. Some are limited during offline, shown in each table below as "✓" and "-".

◇ File

✓ : Selectable and editable during file operation

- : Not selectable nor editable or hidden during file operation

Print	Prints the editor items that support printing. Printing is supported for the [Program], [Point], [Shift], [Hand], [Work], [Palset], and parameter editors.	✓
Print Preview	Shows a print preview.	✓
Print Setup	Shows the print setup screen.	✓
Import	Imports various data from the file.	✓
Export	Exports the current controller and data being edited to a file. (Even when connecting multiple robots, data is exported as one file.)	✓
Exit	Exits the target window.	✓



NOTE

If you execute the File Export command while communication with the controller is disconnected, the data that is exported to the file will be the data that was obtained from the controller before disconnection.

◇ Edit

✓ : Selectable and editable during file operation

- : Not selectable nor editable or hidden during file operation

Reload	Reloads data from the controller, and updates the editor.	✓
Save	Saves the edited content to the controller.	✓
Cut	Cuts the data currently selected in the editor.	✓
Copy	Copies the data currently selected in the editor.	✓
Paste	Pastes the copied data to the editor.	✓
Undo	Returns the edited data to its previous state with canceling the operation executed just before.	✓
Redo	Re-executes the operation that was executed just before.	✓

◇ View

✓ : Selectable and editable during file operation

– : Not selectable nor editable or hidden during file operation

Toolbar	Controller Toolbar	Shows/hides the controller toolbar.	–
	Connection Toolbar	Shows/hides the connection toolbar.	–
	Operation Toolbar	Shows/hides the operation toolbar.	–
	Program Toolbar	Shows/hides the program toolbar.	–
	Coordinate System Toolbar	Shows/hides the coordinate system toolbar.	–
	Parameter Toolbar	Shows/hides the parameter toolbar.	–
	System Toolbar	Shows/hides the system toolbar.	–
	Data Transfer Toolbar	Shows/hides the data transfer toolbar.	–
	Monitor Toolbar	Shows/hides the monitor toolbar.	–
	Edit Toolbar	Shows/hides the edit toolbar.	✓
	Emulator Toolbar	Shows/hides the emulator toolbar.	–
	Small Size Icons	Shows small icons on the toolbar.	✓
	Large Size Icons	Shows large icons on the toolbar.	✓
	Show Text Labels	Shows/hides text on the toolbar.	✓
Window Tree	Shows/hides the Window Tree.		✓
Status Bar	Shows/hides the status bar.		✓
Controller Monitor	Shows/hides the Controller Monitor pane.		–
Current Position	Shows/hides the Current Position pane.		–
DI	Shows/hides the DI pane.		–
DO	Shows/hides the DO pane.		–
MO	Shows/hides the MO pane.		–
LO	Shows/hides the LO pane.		–
TO	Shows/hides the TO pane.		–
SI	Shows/hides the SI pane.		–
SO	Shows/hides the SO pane.		–
SIW	Shows/hides the SIW pane.		–
SOW	Shows/hides the SOW pane.		–
Variable	Shows/hides the variable pane.		–



NOTE

When you change the toolbar (Icon Size) or (Show Text) setting, the setting also applies to the toolbars of the RCX-Studio 2020 window and of other target windows.

◇ Operation

✓ : Selectable and editable during file operation

– : Not selectable nor editable or hidden during file operation

Servo	Opens the Servo dialog box, allowing you to turn the motor power and servos on/off.	–
Origin (All)	Opens the Origin (All) dialog box, allowing you to perform return-to-origin of all the robots when several robots are controlled.	–
Origin (Selected Robot)	Opens the Origin (Selected Robot) dialog box, allowing you to perform return-to-origin of robot by robot.	–
Jog	Shows/hides the Jog pane.	–
Point Trace	Opens the Point Trace dialog box, allowing you to perform point trace.	–

◇ Program

✓ : Selectable and editable during file operation

– : Not selectable nor editable or hidden during file operation

New Program	Opens a new program editor.	✓
Syntax Check	Shows/hides the Syntax Check pane.	✓
Directory	Shows the program directory dialog box, allowing you to edit the program's attributes.	–
Task	Shows/hides the Task pane.	–
Break Point	Shows/hides the Break Point pane.	–
Input/Output	Shows/hides the Input/Output pane.	–
Find Result	Shows/hides the Find Result pane.	✓
Sequence Execution Flag	Shows the Sequence Execution Flag dialog box.	–

◇ Coordinate System

✓ : Selectable and editable during file operation

– : Not selectable nor editable or hidden during file operation

Point	Opens the Point editor.	✓
Standard Coordinate	Opens the Standard Coordinate setup wizard.	–
Shift	Opens the Shift editor.	✓
Hand	Opens the Hand editor.	✓
Work	Opens the Work editor.	✓
Pallet	Opens the Pallet editor.	✓
Area Check Output	Opens the Area Check Output editor.	✓
Camera Calibration *	Opens the Camera Calibration editor.	✓
Conveyor Calibration *	Opens the Conveyor Calibration editor.	✓

* Menu items are displayed during controller operation when a controller is connected actually. Menu items are displayed during file operation when there is data.

◇ Parameter

✓ : Selectable and editable during file operation

– : Not selectable nor editable or hidden during file operation

Controller	Opens the Controller Parameter editor.	–
Robot	Opens the Robot Parameter editor.	–
Axis	Opens the Axis Parameter editor.	–
Driver	Opens the Driver Parameter editor.	–
IO	Opens the IO Parameter editor.	–
Option Board Enable	Opens the Option Board Enable Parameter editor.	–
Option Board Parameter	Opens the Option Board Parameter editor.	–
General Ethernet Port	Opens the General Ethernet Port editor.	✓
Camera *	Opens the Camera Parameter editor.	✓
Light *	Opens the Light Parameter editor	✓
Tracking	Opens the Tracking Parameter editor	✓
Double-carrier	Opens the Double-carrier Parameter editor.	✓
Real-time output	Opens the Real-time Parameter editor	✓
PC Vision	Opens the PC Vision Parameter editor	✓

* Menu items are displayed during controller operation when a controller is connected actually. Menu items are displayed during file operation when there is data.

◇ System

✓ : Selectable and editable during file operation

– : Not selectable nor editable or hidden during file operation

Properties	Shows the controller version and other information.	–
Check	Displays any alarms that have occurred.	–
Alarm History	Shows the history of any alarms that have occurred.	–
Generation	Assigns drivers and specifies the generation of robots and axes.	–
Initialize	Executes initialization.	–
Communication Setting	Edits the settings.	–
Access Level	Changes the access level.	–

◇ Tool

✓ : Selectable and editable during file operation

– : Not selectable nor editable or hidden during file operation

Controller Backup	Transfers data from the controller to the PC.	–
Controller Restore	Transfers data from the PC to the controller.	–
Data Difference	Compares data between the controllers and files.	✓
Cycletime Calculator	Calculates the cycletime.	✓
Option	Specifies options.	✓

◇ Window

✓ : Selectable and editable during file operation

– : Not selectable nor editable or hidden during file operation

Cascade	Displays the currently-running editors overlapping in the edit area.	✓
Tile Horizontally	Displays the currently-running editors tiled horizontally in the edit area.	✓
Tile Vertically	Displays the currently-running editors tiled vertically in the edit area.	✓
Dock All Panes	Docks all panes that are currently floating.	✓

◇ Help

✓ : Selectable and editable during file operation

– : Not selectable nor editable or hidden during file operation

Help	Displays the Help screen.	✓
About RCX-Studio 2020	Displays the software version of RCX-Studio 2020.	✓

Toolbar

The toolbar contains buttons for executing frequently-used menu commands. You can use a command in the main menu to switch the toolbar between visible and hidden.

Window Tree

The Window Tree shows the functions organized in a tree structure.

Controller monitor pane

The status of the controller is shown here in real time.

Jog pane

Here you can move the axes that are connected to the robot.

Current position pane

The current position of the axes connected to the robot are shown here.

Editing area

This area shows the editor and allows you to edit.

Status bar

The status bar shows explanation of the functions that the mouse is pointing and the status of special keys

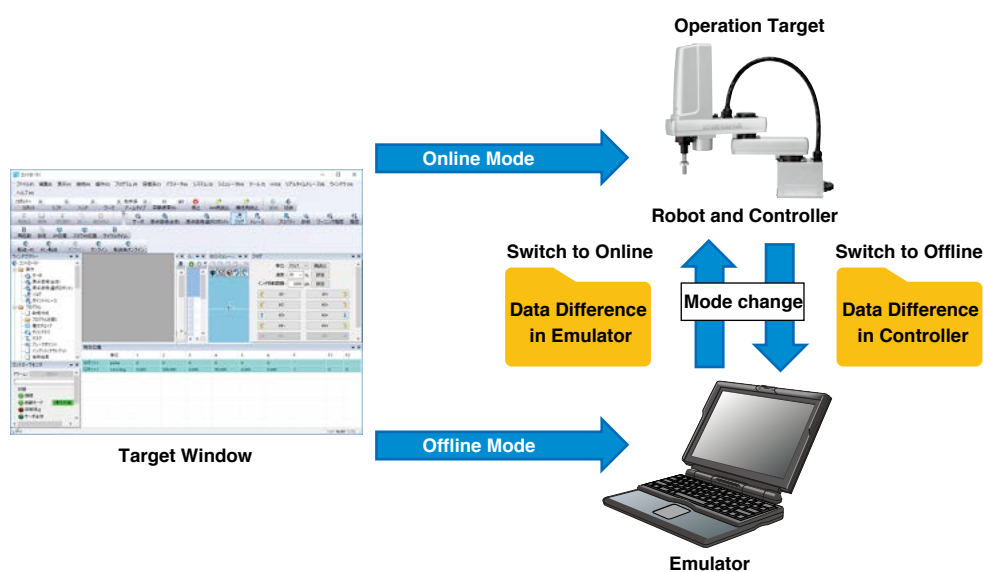
4. Control operation

RCX-Studio 2020 allows you to remotely control robots connected to the controller and monitor their operating status from the PC. The following sections describe how to connect controllers to RCX-Studio 2020 as well as how to control the robot operation.

4.1 Overview of operation

RCX-Studio 2020 performs a connection to an actual controller in the online mode and a connection to an emulator (virtual controller) in the offline mode. By changing two modes, the operation or data edit is performed.

The data change that is performed for each target can be applied to other unit as a difference. A safe and efficient program can be created or the layout can be investigated.



■ Online mode

When an actual controller is connected, the data inside the controller can be edited directly or the controller status can be monitored. In addition, the robot can also be operated by performing the manual operation or executing the program.

■ Offline mode

A program can be created or executed for an emulator (virtual controller). In addition, use of the 3D emulator makes it possible to verify the program or investigate the layout.



WARNING

When operating a robot, make sure that all workers are outside the safety fence, and then the emergency stop is ready to input at any time.

Target data can be imported/exported in the "Controller target data (ctd file)" format. The controller target data includes the settings of the emulator and 3D simulator in addition to the communication settings with the controller, so you can back up and duplicate the simulated environment.

Refer to "4.8 Importing/Exporting target" for how to import/export controller target data.

4.2 Adding a controller

To perform the operation or edit with an actual controller or emulator connected, a controller needs to be added. This section describes how to connect an actual controller and add a controller.

Step 1 Connect the PC to the robot controller.

Using a dedicated communication cable (RS-232C) or an Ethernet cable (category 5 or higher), connect the PC to the robot controller. For details on how to make connections, refer to the manual of the robot controller.

Step 2 Start RCX-Studio 2020.

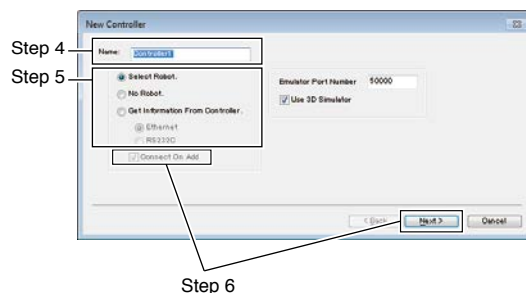
The RCX-Studio 2020 window opens.

Step 3 Select [New Controller] from the [Controller] menu.

The (New Controller) dialog box appears.

TIP

You can get the same result by clicking (New Controller) on the toolbar, or by double-clicking (Target Tree)-(Controller)-(New Controller).



Step 4 Enter the display name.

Enter the name to display in the target tree and in the title bar of the target window.

TIP

The name you specify here is not registered to the robot controller. It is used only to identify that controller within RCX-Studio 2020.

Step 5 Select [Get Information From Controller] from the Data Creation Method.

For the communication method to connect to a controller, select Ethernet or RS232C.

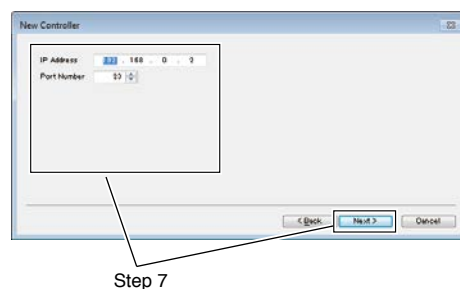
Step 6 Select [Connect On Add], and then click [Next].

If the (Connect On Add) check box is selected, the controller will be connected after it is added to the target tree.

If this is not selected, the controller will only be added to the target tree, and will not be connected. This is selected by default.

Step 7 Make the communication settings, and then click [Next].

To connect to the controller via RS-232C or Ethernet, select the targeting tab and the communication settings from the list boxes. Make sure that the communication settings are the same as that of the controller. For details on how to verify the communication settings and IP address of the robot controller, refer to the manual of the robot controller.



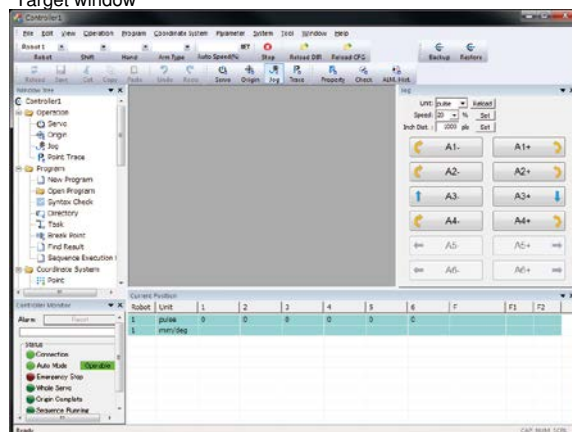
Step 8 Connect to the controller.

Click (OK). The (Finish) dialog box appears. Click (Finish) to add the controller to the target tree.

If the (Connect On Add) setting is selected, the PC connects to the controller. If the connection is successful, the target window appears.

If the connection is unsuccessful, an error message "Connection error" or "Failed to connect" appears. Check the communication settings and the state of the communication cable.

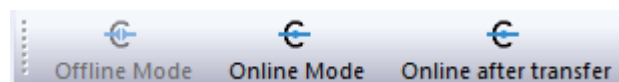
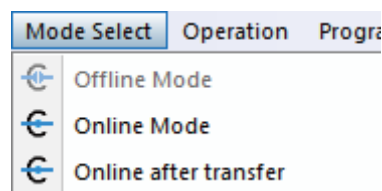
Target window



■ To change the controller mode

To change the controller mode, there are two procedures as described below.

- From [Mode Select] on the main menu in the Target window, select [Offline], [Online], or [Online After Transfer].
- Click [Offline], [Online], or [Online After Transfer] on the toolbar in the Target window.



TIP

Select the controller whose mode you want to change in the target tree in the Target window of the RCX-Studio 2020 window, and then change the controller mode using the operation shown below.

- Select (Controller) - (Online) or (Offline) on the main menu.
- Click (Online) or (Offline) on the toolbar.
- Press the right mouse button, and then select (Online) or (Offline) from the context menu.

■ If a connection error occurs in the online mode

If due to a broken cable or other reason, a connection error occurs while communicating with the robot controller, the reconnection confirmation message will appear. Select [Yes] to perform the reconnection. When [No] is selected, the file save confirmation message appears. Save the file as required. After that, the mode is changed to the offline mode.

If a connection error occurs continuously, check the connection status of the communication cable, and then check the communication settings of the PC and robot controller.

4.3 Servo on

In order to return the robot to origin position or operate it, the power and servo must be turned on.

Step 1 Open the [Servo] dialog box.

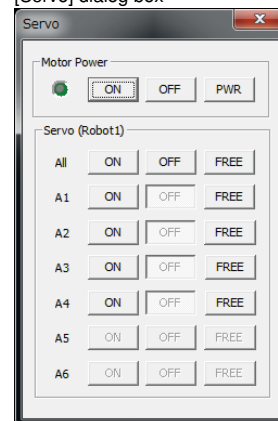
Use any of the following methods to open the dialog box.

- On the main menu, click (Operation)-(Servo).
- Double-click (Servo) in the Window Tree.

Step 2 Turn on the motor power.

Click Motor Power (ON) or (PWR). The motor power turns on, and the motor power lamp is lit.

[Servo] dialog box



■ The difference between Motor Power [ON] and [PWR]

When clicking [ON], the execution confirmation message will appear.

By executing, the motor power turns on and simultaneously all axes of the robot will be in the servo-on status.

When clicking [PWR], although the motor power turns on, the robot's axes will not be in the servo-on status.

Step 3 Turn the servo of each axis on.



CAUTION

When clicking servo (FREE), the servo turns free, and the axis mechanical brake and the dynamic brake are released. As the mechanical brake is released, the robot that operates vertically may drop. Ensure safety before executing this operation.

Servo (ON), (OFF), and (FREE) buttons are provided for All axes and for individual axis A1 to A6. These buttons are enabled for each axis that is connected to the robot.

■ About Servo [ON]

When clicking Servo [ON] of each axis, the corresponding axis changes to servo-on status. When clicking Servo [ON] of [All] axes, all the axes connecting to the robot change to servo-on status.

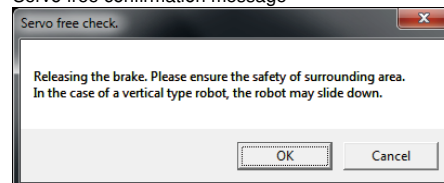
■ About Servo [OFF]

When clicking Servo [OFF], the corresponding axis changes to the servo-off status.

■ About Servo [FREE]

When clicking Servo [FREE], a confirming message window appears. To put the servo-free status, click [OK].

Servo free confirmation message



NOTE

To change the servo status of a different robot, change the robot selection in the controller toolbar.

4.4 Return-to-origin

Return-to-origin of the robot must be performed for acquiring and using the robot position. The following section describes how to return the robot to origin position.

Step 1 Open the [Origin] dialog box.

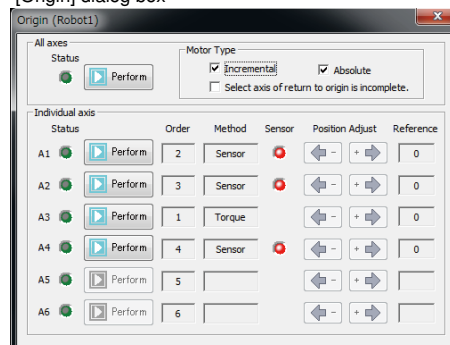
Use either of the following methods to open the dialog box.

- On the main menu, click (Operation)-(Origin).
- In the Window Tree, double-click (Origin).

Step 2 Press [Perform] for the axis to perform return-to-origin.

When clicking (Perform) for the axis, it will begin the return-to-origin operation. During return-to-origin execution, (Perform) button changes to (Stop) button. To interrupt the return-to-origin operation, press (Stop). When return-to-origin is completed, the status lamp is lit.

[Origin] dialog box



■ In the case of Mark method

The Position Adjust buttons are valid. Press the Position Adjust [-] or [+] buttons to move to a position where absolute reset is possible, and then press [Perform].

■ Performing return-to-origin for all axes

To perform return-to-origin for all axes, press [Perform] for All axes.

Use the following three check boxes to select the motor type(s).

- Incremental
Perform return-to-origin for all the incremental axes.
- Absolute
Perform return-to-origin for all the absolute axes.
- Select axis of return to origin is incomplete
Perform return-to-origin only for the axes that are origin incomplete.
If [Select axis of return to origin is incomplete] is selected, the [Incremental] and [Absolute] settings are invalid.

■ About the Status lamps

These are unlit if return-to-origin is incomplete for the corresponding axis, and are lit if return-to-origin is complete. For "All axes," the Status lamp is lit when return-to-origin is complete for all axes connected to the robot; it is unlit if there is even one axis for which return-to-origin is incomplete.



NOTE

- To perform return-to-origin for another robot, change the robot selection in the controller toolbar.
- The return-to-origin order can be changed by editing the robot parameter settings.
- The return-to-origin method can be changed by editing the axis parameters. For details, refer to "7.6 Editing parameters."

4.5 Moving the robot in inching or jog mode

This section describes how to move the robot in inching or jog mode.

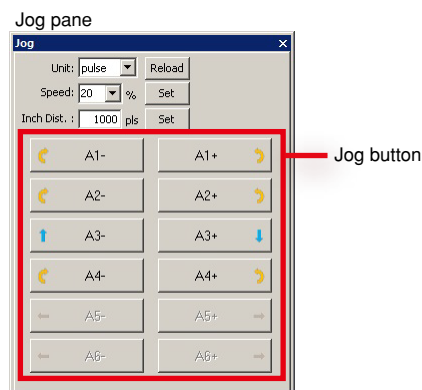
Step 1 Open the Jog pane.

If the Jog pane is not shown in the target window, use any of the following methods to display it.

- On the main menu, click (Operation)-(Jog).
- On the operation toolbar, press (Jog).
- In the Window Tree, double-click (Jog).

TIP

If the Jog pane is not seen, it might be displayed outside the screen. In the main menu, try executing (Window)-(Dock All Panes).



Step 2 Choose the unit from the list box.

One of three units; (pulse), (mm)*, or (tool)*, can be chosen for your situation.

*"Std. coord. doesn't exist" message will be shown when connecting to the emulator and selecting SCARA robot. On the main menu, click (Coordinate System) - (Standard Coordinate) and set the standard coordinate on the displayed wizard.

Step 3 Set the speed of inching movement or jog movement.

Speed can be set by either choosing 1, 2, 5, 10, 20, 50, or 100 from the list or manually entering a value between 1 and 100 into the box.

If the value differs from the controller's setting, the list box's background color turns red.

Click the (Set) button located at the right of the (Speed) list box to apply the setting to the controller.

Step 4 Set the inching movement distance (Inch Dist.).

Inching distance can be set by entering directly into the box either number between 1 and 10000 in "pulse" unit or between 0.001 and 10.000 in "mm" or "tool" unit. If the value differs from the controller's setting, the edit box's background color turns red.

Click the (Set) button located at the right of the (Inch Dist.) edit box to apply the setting to the controller.

TIP

If you click the (Reload) button, the speed and inching movement distance values are returned to the values of the controller.

Step 5 Operate the robot.



WARNING

The robot will operate by pressing the jog buttons. Do not enter the robot movement range; doing so is hazardous.

Each time clicking the jog button, the robot moves a distance set as the inching moving distance in the respective direction.

If you hold down a jog button, the robot keeps moving as long as you continue holding down it. When releasing the button, the robot slows down and stops. The robot also slows down and stops when it reaches the soft limit in the respective directions.

4.6 Moving the robot to a specified point (Point Trace)

This section explains how to move the robot to a point that is set beforehand.

Step 1 Open the Point Trace dialog box.

Use any of the following methods to open the dialog box.

- On the main menu, click (Operation)-(Point Trace).
- In the (Operation Toolbar), click (Point Trace).
- In the Window Tree, double-click (Point Trace).
- In the Point editor, click (Trace).

[Point Trace] dialog box



Step 2 Select the trace type and the motion type.

The trace type can be specified as (Absolute), (Incremental), or (Tool Coordinate). The motion type can be specified as (Point To Point), (Linear Interpolation), or (Circular Interpolation). (Circular Interpolation can be selected only if the trace type is "Absolute.")

Step 3 Set the trace option speed.

The trace operation speed is the product of the movement speed and the option speed. If the option speed is not selected, the robot moves at the movement speed.

TIP

The movement speed setting can be viewed and edited in (Auto Speed (%)) of the control toolbar.

Step 4 Select a point, or select position data entry.

If you selected (Select Point) in the point trace settings, select the point number to execute point trace. When selecting (Enter Position), you can directly input a movement destination for the robot into the point data.

Step 5 Click [Trace] button to begin trace movement.



WARNING

The robot will operate by pressing the jog buttons. Do not enter the robot movement range; doing so is hazardous.

During trace movement (Trace) button changes to (Stop) button. To stop the movement while it is in progress, click (Stop) button.

■ Selecting multiple points and executing trace

Select the [Multiple Select] check box, add points to the multiple select list, and click [Multiple Trace]); trace movement occurs through the selected points.

If [CONT] check box is selected, the option is enabled. For details on CONT, refer to the programming manual of the robot controller.

■ Arch option

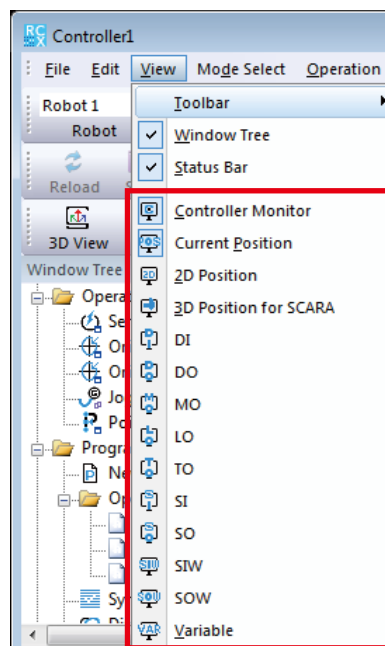
If the trace type is [Absolute] and the motion type is [Point To Point], the arch option can be specified.

For details on the arch option, refer to the programming manual of the robot controller.

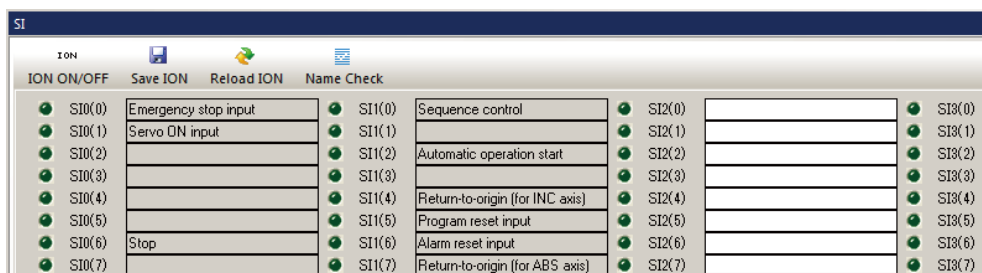
4.7 Monitor function

Follow the procedures below to specify / refer to the current conditions of the internal controller.

Step 1 Select the item to monitor.



Step 2 The pane to monitor will be shown.

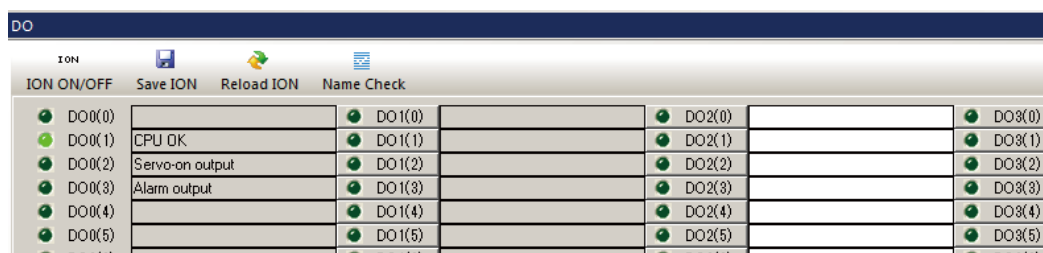


Step 3 Each output signal is turned ON/OFF by clicking each item on the [DO/MO/LO/TO/SO] screen.

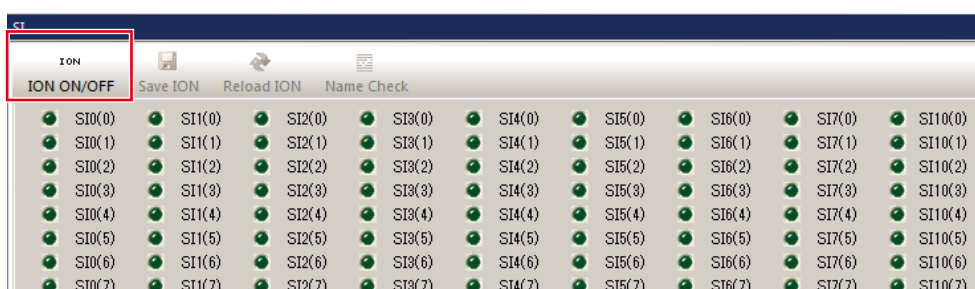


NOTE

It is not applicable to operate dedicated outputs (such as return-to-origin).



Step 4 IO (Input/Output) name can be switched by clicking "ION ON/OFF" button for I/O with the function. (Refer to 4.6.1 "IO name function".)



4.7.1 IO name function

Each signal for DI/DO and SI/SO can be named. (IO name function)
This name can be used in a program or referred to with the programming box.
Procedures of editing and saving the name are described in this section.

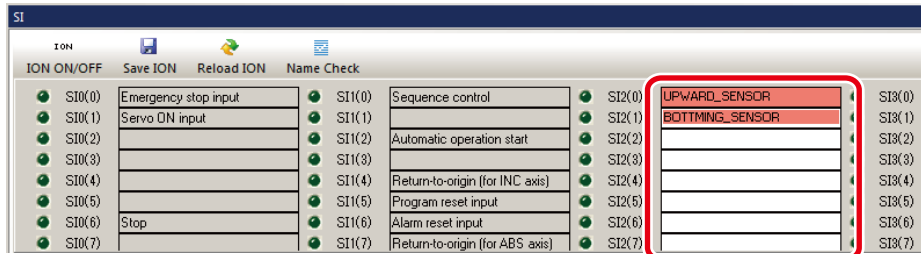


NOTE

Refer to RCX3 series User's Manual Chapter 10 "3. I/O status display" for referring to names with programming box.

Step 1 Enter a name in any empty box of DI/DO and SI/SO.

The box will be changed to red.



CAUTION

The IO names have not been transferred to the controller yet in this state.



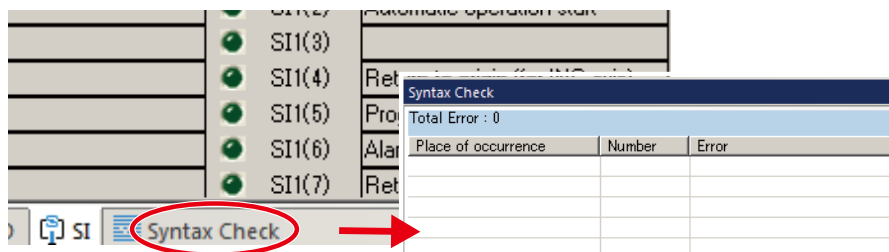
NOTE

For IO names;

- Available characters: Alphanumeric and _ (under bar)
- Number of characters: Maximum 16

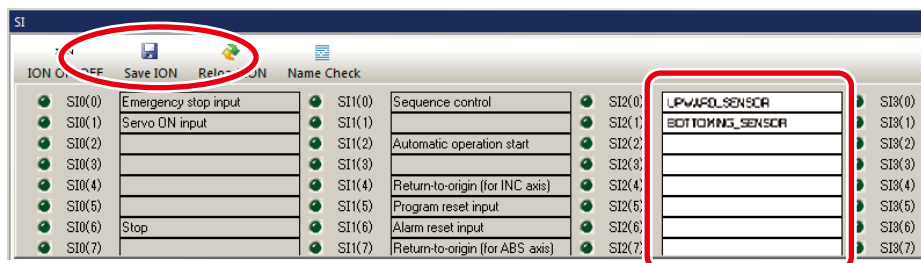
Step 2 Confirm that no error is occurring in the pane by "Syntax Check" of "Name Check".

If an error occurs, re-enter the name while referring to the NOTE of 1.



Step 3 Click "Save ION" button.

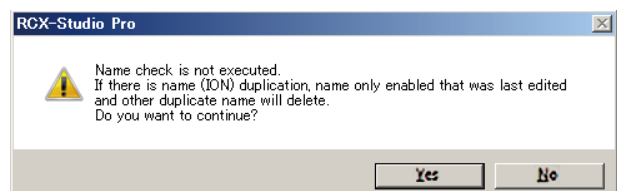
The box will return to white once the IO name is transferred to the controller.



NOTE

Click "Reload ION" button to read out the IO name saved in the controller.

The message window on the right will be shown if "Save ION" is clicked without executing "Syntax Check". Select Yes/No.



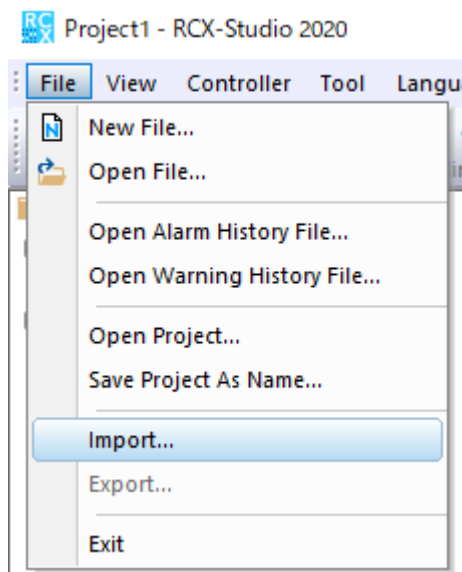
4.8 Importing/Exporting target

This section describes how to import/export controller target data.

4.8.1 Import

Step 1 *Open the import screen.*

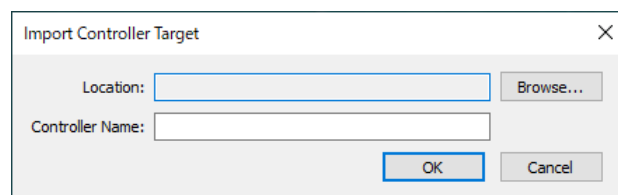
Select (File)-(Import).



Step 2 *Select the controller target data to import.*

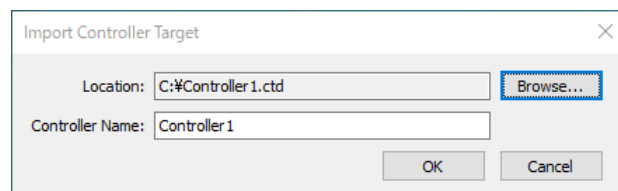
Click the (Browse) button on the import screen to display the file selection screen.

In the file selection screen, select the controller target data (ctd file) to be imported.



Step 3 *Sets the name of the controller target.*

Enter the name of the controller target created by importing in (Controller Name), and press (OK).

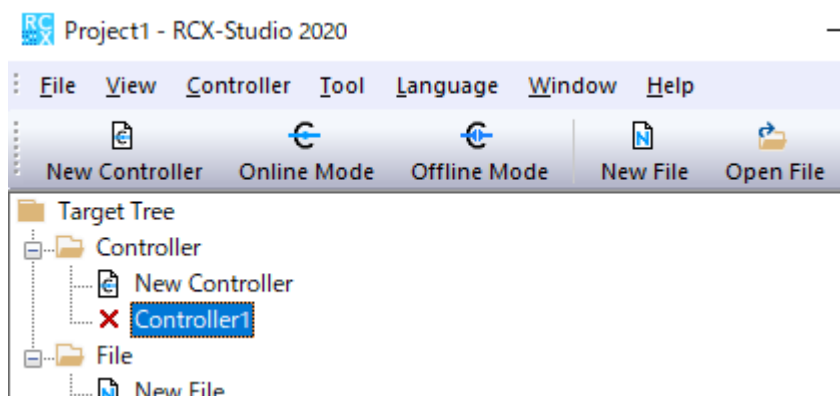


NOTE

- You cannot set the same name as a target that already exists in the target tree.
- If you select the exported data in a different version, the return to origin state or program execution state will be reset.

Step 4 *Confirm the controller target.*

Confirm that the controller target has been added to the target tree.



4.8.2 Export

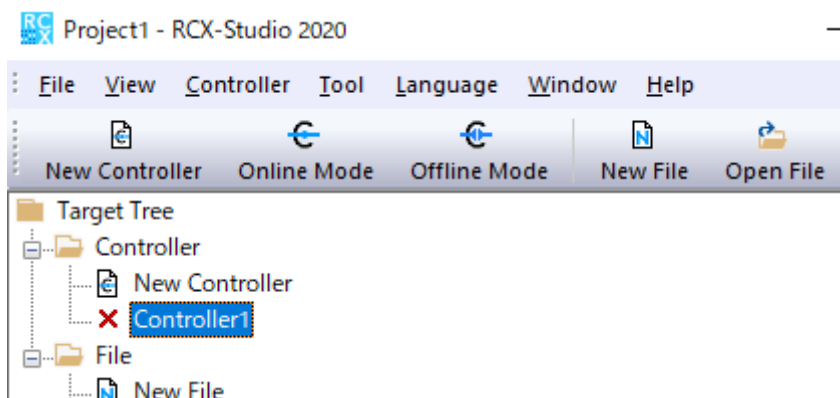
Step 1 Select the controller target to export.

Select the controller target you want to export in the target tree.



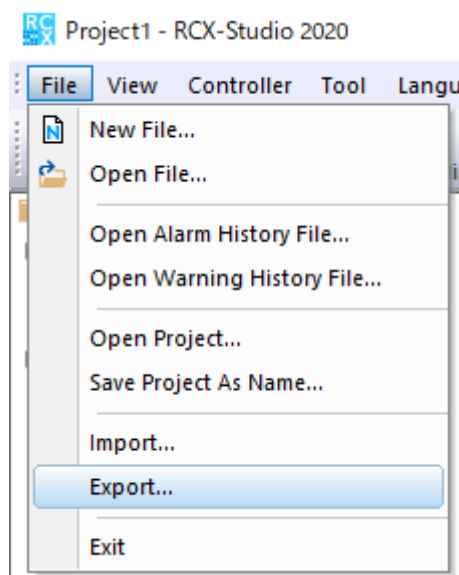
NOTE

The target to be exported must be closed.



Step 2 Open the export screen.

Select (File)-(Export).



Step 3 Save as controller target data.

Specify the file name and save location, and then press the (Save) button.

5. File operation

While you can edit a controller data file obtained via data transfer or edit a newly created data file. Data edited during can be sent to the controller.

5.1 Creating a new data file

This section explains how to create a new data file.

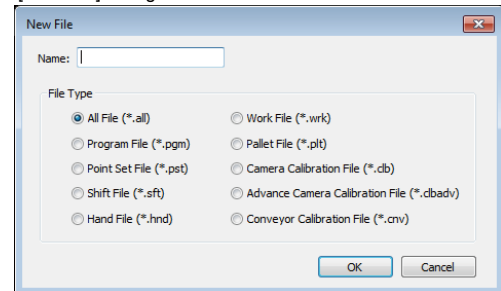
Step 1 On the [File] menu, click [New File].

The (New File) dialog box appears.

TIP

The same result is gotten by clicking (New File) on the toolbar, or by double-clicking (Target Tree)-(File)-(New File).

[New File] dialog box



Step 2 Enter a name.

Enter the name to be saved.

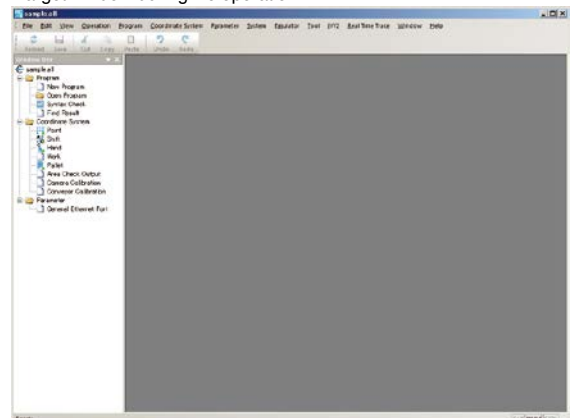
Step 3 Select the file type.

The editable content differs according to the file type that you select. In the "All File", any type of files can be edited.

Step 4 Open the target window.

When clicking (OK), the file is added to the Target Tree, and the target window appears.

Target window during file operation



NOTE

The file is not created at opening the target window. After editing the data, the file is actually created by saving from the (Save As) dialog box. The edited content is saved in the file that is created. If closing the target window without creating the file, the corresponding file is deleted from the Target Tree of the RCX-Studio 2020 window.

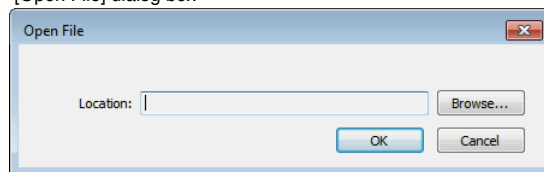
5.2 Opening a data file

This section explains how to open a data file that was saved by transferring controller data or by saving a newly created data file.

Step 1 On the [File] menu, click [Open File].

The (Open File) dialog box appears.

[Open File] dialog box



TIP

The same result is gotten by clicking (Open File) on the toolbar, or by double-clicking (Target Tree)-(File)-(Open File).

Step 2 Select the controller type.

Select the robot controller from the list box.

Step 3 Select the file that you want to open.

Click (Browse) and select the file to open.

The following table shows the file extensions that you can open and the content that you can edit.

■ Editable items for each file type ✓: Editable -: Non-editable

Extension	Program	Point	Shift	Hand	Work	Pallet	Area Check Output	Parameter	General Ethernet Port
.all *1	✓	✓	✓	✓	✓	✓	✓	✓	✓
.all *2	✓	✓	✓	✓	✓	✓	✓	-	✓
.pgm	✓	-	-	-	-	-	-	-	-
.pst	-	✓	-	-	-	-	-	-	-
.pnt	-	✓	-	-	-	-	-	-	-
.pnm	-	✓	-	-	-	-	-	-	-
.pcm	-	✓	-	-	-	-	-	-	-
.prm	-	-	-	-	-	-	-	✓	-
.sft	-	-	✓	-	-	-	-	-	-
.hnd	-	-	-	✓	-	-	-	-	-
.plt	-	-	-	-	-	✓	-	-	-

*1) An .all file obtained by transferring controller data

*2) An .all file created as a new file and saved

Step 4 Open the target window.

When clicking (OK) in the (Open File) dialog box, the file is added to the Target Tree and the target window opens. Edit the data in the target window and save it into the file.

Target window during file operation

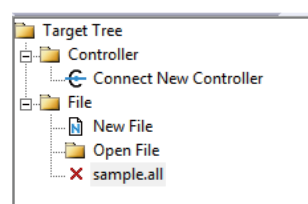


5.3 Deleting a file from the Target Tree

This section describes how to delete a file from the Target Tree of the RCX-Studio 2020 window.

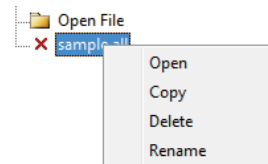
Step 1 *In the Target Tree, select the file to delete.*

In the RCX-Studio 2020 window, select the file to delete.



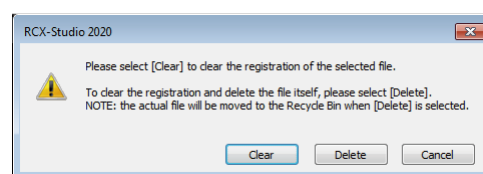
Step 2 *Select "delete".*

Select the file and right-click to delete.



Step 3 *Confirmation message appears.*

A message confirming that you want to delete the data file will appear.



CAUTION

When you select **(Delete)** in the delete confirmation message, the original data file is also deleted. The deleted data file is moved to Recycle Bin.

The following actions are in response to the delete confirmation message.

Clear	The selected file registration is cancelled from the Target Tree.(The original data file is not deleted.)
Delete	Deletes the selected file from the target tree and moves the original data file to Recycle Bin.
Cancel	No action.

6. Data transfer

RCX-Studio 2020 includes function for transferring data to any desired location. Controller data can be saved into PC and the saved data can then be restored to the controller.

6.1 Transferring data from controller to PC

This section explains how to transfer controller data to the PC and save it as a file in the location you specify. To transfer data from a currently-connected controller to the PC, execute this procedure from the target window of the corresponding controller. To transfer data from an unconnected controller to the PC, execute this procedure from the RCX-Studio 2020 window.

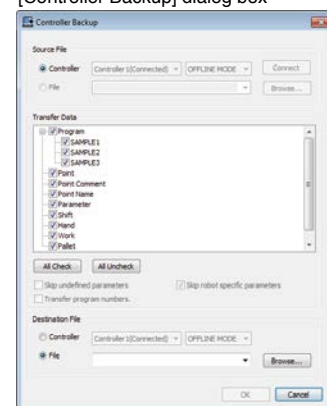
■ Transferring data from a connected controller to the PC

Step 1 Open the [Controller Backup] dialog box.

Use any of the following methods to open the (Controller Backup) dialog box.

- On the main menu of the target window, click (Tool)- (Controller Backup).
- In the data transfer toolbar of the target window, click (Backup).

[Controller Backup] dialog box



Step 2 Select the data that you want to transfer to the PC.

In the (Transfer Data) list, select the check box for each type of data to transfer to the PC. (All items are selected by default.)

When undefined parameters are NOT loaded;
Check (Skip undefined parameters).

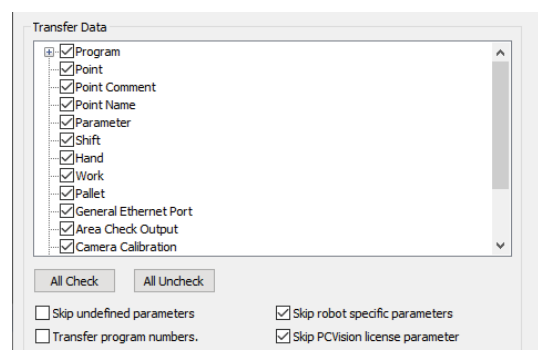
When model specific parameters are NOT loaded;
(including generation data and offset pulse)
Check (Skip robot specific parameters).

When program numbers are loaded;
Check (Transfer program numbers).

When PCVision license parameter is NOT loaded; Item is selected by default.
Check (Skip PCVision license parameter). The license key is NOT loaded.

If uncheck and the parameter is transferred, the license key will be included in the transfer.

At this time, if the transfer destination is RCX3xx controller, the license key in the controller will be overwritten.



TIP

The transfer data list shows only the items for which data is saved in the controller. Items for which no data is saved are not shown in the transfer data list.

Step 3 Specify the transfer destination.

Specify the destination to which the controller data will be transferred.
As the transfer destination you can choose either (Controller) or (File).

■ Transfer destination: Controller

Choose this if you want to transfer data to another controller. Data can be transferred only to an unconnected controller. For a currently-connected controller, [OK] is unavailable and the data cannot be transferred.

■ Transfer destination: File

Choose this for transferring.
When you click [Browse], the [Save As] dialog box appears. Select the transfer-destination folder, and enter a file name for the destination file in the [File name] box.



NOTE

The file name extension (*.all) is automatically added to the file that is transferred from the controller. There is no need to select the file name extension from the (Save as type) dropdown list in the (Save As) dialog box.

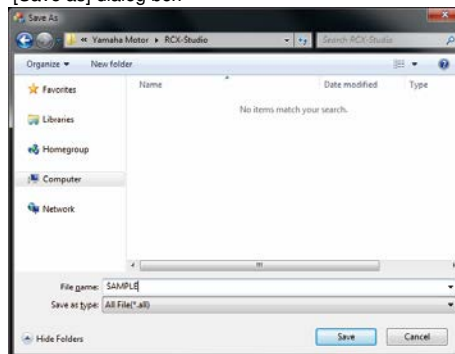
Unconnected controller (data transfer is possible)

☒ Controller Controller 1 OFFLINE MODE

Connected controller (data transfer is not possible)

☒ Controller Controller 1(Connected) OFFLINE MODE

[Save as] dialog box



Step 4 Start transferring the controller data.

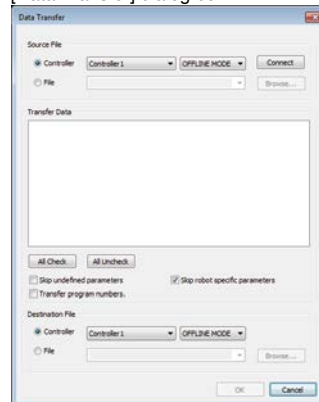
In the (Controller Backup) dialog box, click (OK). Data transfer begins.
When transfer is complete, a transfer completion message appears. Click (OK).

■ Transferring data from an unconnected controller to the PC

Step 1 Open the [Data Transfer] dialog box.

On the main menu of the RCX-Studio 2020 window, click (Tool)-(Data Transfer) to open the (Data Transfer) dialog box.

[Data Transfer] dialog box



Step 2 Select the transfer-source controller.

In (Controller) of (Source File), select the controller from which you want to transfer data to the PC.
Only unconnected controllers can be selected as the transfer source. The (Connect) button is invalid for controllers that are currently connected.

Unconnected controller (data transfer is possible)

☒ Controller Controller 1 OFFLINE MODE Connect

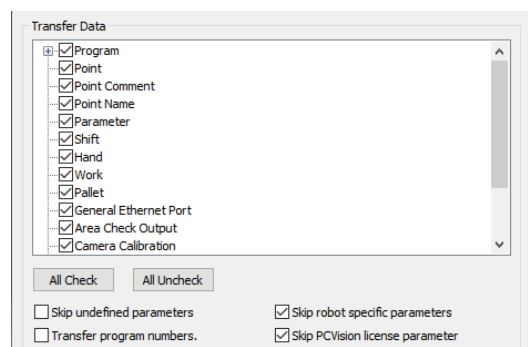
Connected controller (data transfer is not possible)

☒ Controller Controller 1(Connected) OFFLINE MODE Connect

Step 3 Get a list of the transfer data.

Click the (Connect) button located beside the transfer-source controller. A list of the transfer data is obtained from the controller and displayed in the (Transfer Data) area.

If connection with the controller was unsuccessful, the error message "Error: Timeout" or "Failed to connect" appears. If one of these appears, check the communication settings and the connection state of the cable.



Step 4 *Select the data to transfer to the PC.*

In the (Transfer Data) list, select the check box for each type of data to transfer to the PC.
(All items are selected by default.)

When undefined parameters are NOT loaded;

Check (Skip undefined parameters).

When model specific parameters are NOT loaded; (including generation data and offset pulse)

Check (Skip robot specific parameters).

When program numbers are loaded;

Check (Transfer program numbers).

When PCVision license parameter is NOT loaded; Item is selected by default.

Check (Skip PCVision license parameter). The license key is NOT loaded.

If uncheck and the parameter is transferred, the license key will be included in the transfer.

At this time, if the transfer destination is RCX3xx controller, the license key in the controller will be overwritten.

TIP

The transfer data list shows only the items for which data is saved in the controller. Items for which no data is saved are not shown in the transfer data list.

Step 5 *Specify the transfer destination for the controller data.*

As the transfer destination, you can choose (Controller) or (File).

When (Controller) is chosen, (online mode) (actual machine) or (offline mode) (emulator) can be selected.

■ Transfer destination: Controller

Choose this to transfer data to another controller.

Data can be transferred only to an unconnected controller. For a currently-connected controller, [OK] is invalid and the data cannot be transferred.

Unconnected controller (data transfer is possible)

☒ Controller Controller1 OFFLINE MODE

Connected controller (data transfer is not possible)

☒ Controller Controller1(Connected) OFFLINE MODE

■ Transfer destination: File

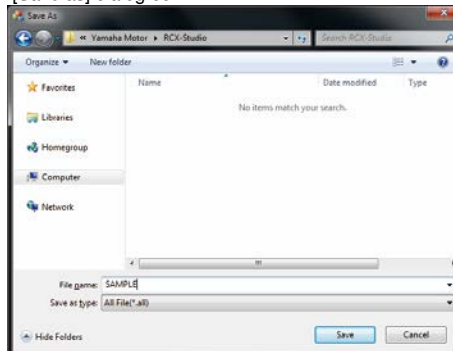
Choose this to transfer data to a file.

When you click [Browse], the [Save As] dialog box appears. Select the transfer-destination folder, and enter a file name for the destination file in the [File name] box.

**NOTE**

The file name extension (*.all) is automatically added to the file that is transferred from the controller. There is no need to select the file name extension from the (Save as type) dropdown list in the (Save As) dialog box.

[Save as] dialog box

**Step 6** *Start transferring the controller data.*

In the (Controller Backup) dialog box, click (OK). Data transfer begins, and the following message appears.

When transfer is complete, a transfer completion message appears. Click (OK).

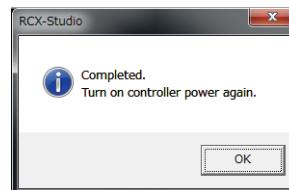
6.2 Transferring data from PC to controller

This section explains how to transfer data from another controller or from a file on the PC to a controller. To transfer data to a currently-connected controller, execute this procedure from the target window of that controller. To transfer data to an unconnected controller, execute this procedure from the RCX-Studio 2020 window.



CAUTION

If you transferred data that requires the power to be cycled, the transfer completion message shown at right appears. Click (OK), and turn the controller's power off and then on again.



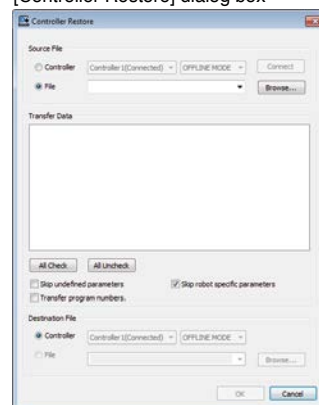
■ Transferring data to a connected controller

Step 1 Open the [Controller Restore] dialog box.

Use any of the following methods to open the (Controller Restore) dialog box.

- On the main menu of the target window, click (Tool)-(Controller Restore).
- In the data transfer toolbar of the target window, click (Restore).

[Controller Restore] dialog box



Step 2 Specify the transfer source.

Specify the source from which the data is transferred.

As the transfer source you can choose either (Controller) or (File).

When (Controller) is chosen, (online mode) (actual machine) or (offline mode) (emulator) can be selected.

■ Transfer source: Controller

Choose this to transfer data from another controller.

Only an unconnected controller can transfer data.

A currently-connected controller cannot transfer data. Select an unconnected controller, and press [Connect] to obtain a list of the transfer data.

Unconnected controller (data transfer is possible)



Connected controller (data transfer is not possible)



■ Transfer source: File

Choose this to transfer data from a file on the PC.

When you click [Browse], the [Open] dialog box appears. Select the transfer-source file.

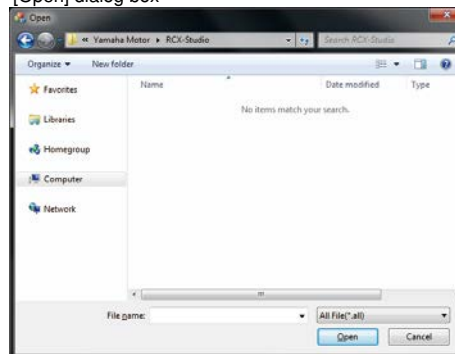


NOTE

Select the file name extension of the transfer-source file from the dropdown list located beside the file name.

The following file name extensions can be selected: (*.dll), (*.pgm), (*.pst), (*.pnt), (*.pnm), (*.pcm), (*.prm), (*.sft), (*.hnd), (*.wrk), (*.plt), (*.clb), (*.clbadv), and (*.cnv).

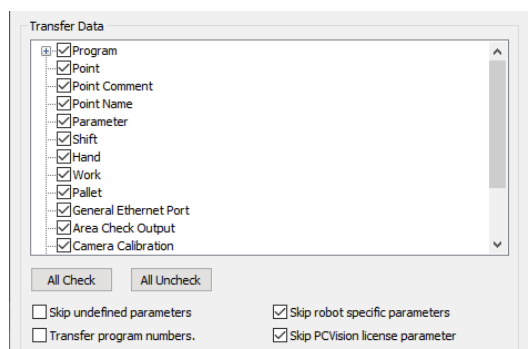
[Open] dialog box



Step 3 *Select the data to transfer to the controller.*

(If the transfer-source is a controller)
a list appears when pressing (Connect).

(If the transfer-source is a file)
a list appears when opening the file.



In the (Transfer Data) list, select the check box for each type of data to transfer to the controller.

When undefined parameters are NOT loaded;

Check (Skip undefined parameters).

When model specific parameters are NOT loaded; (including generation data and offset pulse)

Check (Skip robot specific parameters).

When program numbers are loaded;

Check (Transfer program numbers).

When PCVision license parameter is NOT loaded; Item is selected by default.

Check (Skip PCVision license parameter). The license key is NOT loaded.

If uncheck and the parameter is transferred, the license key will be included in the transfer.

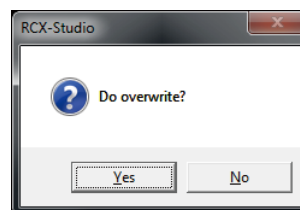
At this time, if the transfer destination is RCX3xx controller, the license key in the controller will be overwritten.

TIP

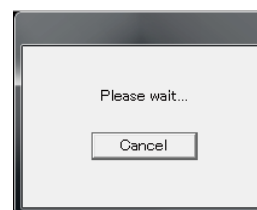
The transfer data list shows only the items for which data is saved in the controller. Items for which no data is saved are not shown in the transfer data list.

Step 4 *Start transferring the data to the controller.*

In the (Controller Restore) dialog box, click (OK). The message "Do overwrite?" appears. Click (Yes) to start transferring. If you decide not to transfer, click (No).



During data transfer, the message shown at right appears.

**NOTE**

To cancel the transfer, click (Cancel).

**CAUTION**

Even if you cancel the transfer, the data that had been transferred to the controller up to the moment you clicked (Cancel) will not return to the state prior to transfer.

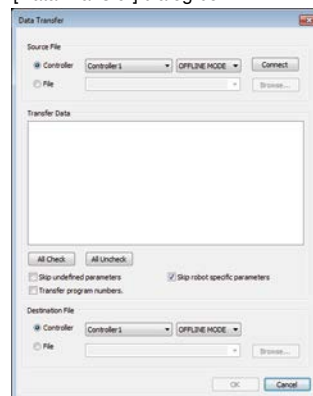
When transfer is complete, a transfer completion message appears. Click (OK).

■ Transferring data to an unconnected controller

Step 1 **Open the [Data Transfer] dialog box.**

On the main menu of the RCX-Studio 2020 window, click (Tool)-(Data Transfer) to open the [Data Transfer] dialog box.

[Data Transfer] dialog box



Step 2 **Specify the source from which the data is transferred.**

As the transfer source you can choose either (Controller) or (File).

When (Controller) is chosen, (online mode) (actual machine) or (offline mode) (emulator) can be selected.

■ Transfer source: Controller

Choose this to transfer data from another controller. Only an unconnected controller can transfer data. Data of a currently-connected controller cannot be transferred. Select an unconnected controller, and press [Connect] to obtain a list of the transfer data.

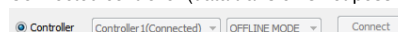
■ Transfer source: File

Choose this to transfer data from a file on the PC. When you click [Browse], the [Open] dialog box appears. Select the transfer-source file.

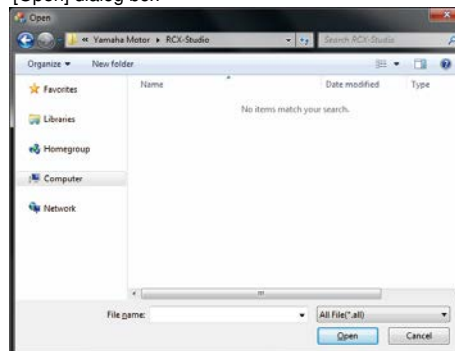
Unconnected controller (data transfer is possible)



Connected controller (data transfer is not possible)



[Open] dialog box



NOTE

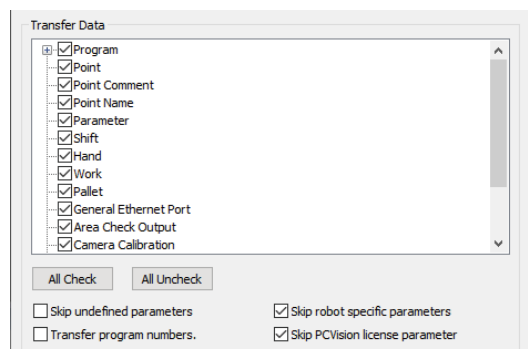
Select the file name extension of the transfer-source file from the dropdown list located beside the file name.

The following file name extensions can be selected: (*.all), (*.pgm), (*.pst), (*.pnt), (*.pnm), (*.pcm), (*.prm), (*.sft), (*.hnd), (*.wrk), (*.plt), (*.clb), (*.clbadv), and (*.cnv).

Step 3 *Select the data to transfer to the controller.*

(If the transfer-source is a controller)
a list appears when pressing (Connect).

(If the transfer-source is a file)
a list appears when opening the file.



In the (Transfer Data) list, select the check box for each type of data to transfer to the controller.

When undefined parameters are NOT loaded; Check [Skip undefined parameters].

When model specific parameters are NOT loaded; (including generation data and offset pulse)
Check (Skip robot specific parameters).

When program numbers are loaded; Check [Transfer program numbers].

When PCVision license parameter is NOT loaded; Item is selected by default.
Check (Skip PCVision license parameter). The license key is NOT loaded.

If uncheck and the parameter is transferred, the license key will be included in the transfer.
At this time, if the transfer destination is RCX3xx controller, the license key in the controller will be overwritten.

TIP

The transfer data list shows only the items for which data is saved in the controller. Items for which no data is saved are not shown in the transfer data list.

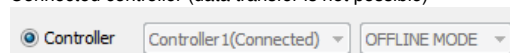
Step 4 *Specify the transfer destination.*

In the controller dropdown list, choose an unconnected controller. If you choose a currently-connected controller, the (OK) button of the (Controller Restore) dialog box is unavailable, and data transfer is not possible.

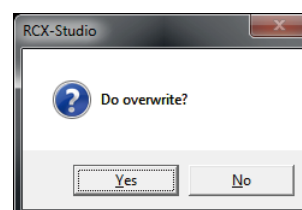
Unconnected controller (data transfer is possible)



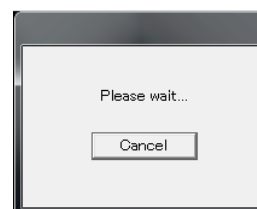
Connected controller (data transfer is not possible)

**Step 5** *Start transferring the data to the controller.*

In the (Controller Restore) dialog box, click (OK). When connection to the controller succeeds, the message "Do overwrite?" appears. Click (Yes) to start transferring. If you decide not to transfer, click (No).



During data transfer, the message shown at right appears.

**NOTE**

If connection to the controller fails, the error message (Failed to connect) appears. If this appears, check the communication settings and the connection state of the cable.

**CAUTION**

Even if you cancel the transfer, the data that had been transferred to the controller up to the moment you clicked (Cancel) will not return to the state prior to transfer.

When transfer is completed, a message indicates completion. Click (OK).

7. Editing controller data and operating

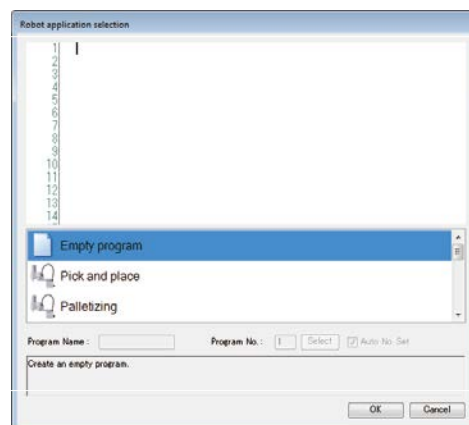
The robot controller's various data (such as program, point, parameter, shift, hand, work, pallet, area output check, camera calibration, conveyor calibration, and general Ethernet port data) can be edited directly on RCX-Studio 2020.

7.1 Creating a new program

Step 1 Open the [Robot application select] dialog box.

Use any of the following methods to display the (Robot application select) dialog box.

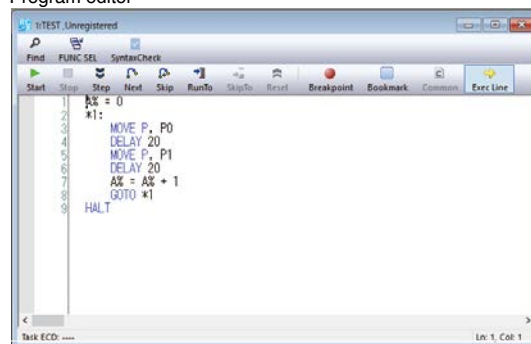
- On the main menu of the target window, click (Program)-(New Program).
- On the (Program Toolbar) of the target window, click (New PGM.).
- In the Window Tree of the target window, double-click (Program)-(New Program).



Step 2 Open a program editor.

Select the (Empty program) from the (Robot application selection) dialog box, and then click (OK) to open a program editor.

Program editor



NOTE

If a program other than (Empty program) is selected in the (Robot application selection) dialog box, the wizard to create a program template can be started. Select the application and follow the messages that appear on the wizard screen to automatically create a program. Use the wizard in accordance with the application.

Step 3 Enter a program into the program editor.

Create a program using the YAMAHA robot language. For details on the YAMAHA robot language, refer to the programming manual of the controller you're using.

Step 4 Save the program.

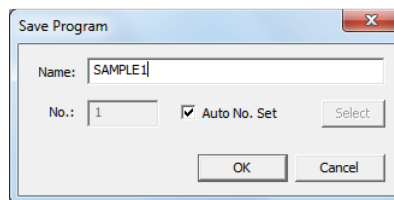
Save the program into the controller directly by either way of the following; click (Edit)-(Save) on the main menu of the target window, or click (Save) in the edit toolbar.



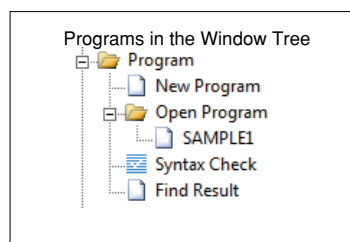
NOTE

If the edit toolbar is not shown in the target window, check whether the main menu (View)-(Toolbar)-(Edit Toolbar) option has a check mark. If not, click (Edit Toolbar). If it already has a check mark, click (Window)-(Dock All Panes).

The (Save Program) dialog box appears when creating a new program or saving a program. Enter a name for the program. The program name can contain single-byte alphanumeric characters and the _ (underscore) character.



If (Auto No. Set) is on, the lowest unused program is automatically assigned. To set a specific program number, clear the (Auto No. Set) check box and click (Select) to select the desired program number. When clicking (OK), the program is saved, and the saved program is added to (Open Program) in the Window Tree.



NOTE

Up to 100 programs can be saved in the controller. If you attempt to save when there are already 100 programs saved in the controller, a "program number over" error appears.

7.2 Editing an existing program

This section explains how to edit a program that was previously saved in the controller.

Step 1 Open the desired program.

In the target window, select a program located under (Window Tree)-(Program)-(Open Program), and double-click it.

Step 2 Enter a program into the program editor.

Create a program using the YAMAHA robot language. For details on the YAMAHA robot language, refer to the programming manual of the controller you're using.

Step 3 Save the program.

Save the program edited using the program editor. Click (Edit)-(Save) in the main menu of the target window, or click (Save) in the edit toolbar. When saving is complete, the "***" at the end of the program editor's title bar disappears.



7.3 Copying or deleting a program

This section explains how to copy or delete a program existing in the controller.

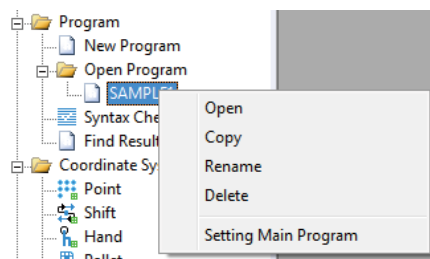
Step 1 Select the program to copy or delete.

In the target window, select the program in (Window Tree)-(Program)-(Open Program).

Step 2 Choose [Delete] or [Copy].

Select a program, right-click to open menu. Click (Copy) to show the (Save Program) dialog box. Enter a program name and click (OK) to copy the program. Click (Delete) to delete the selected program.

* Click (Rename) to rename the selected program, or click (Setting Main Program) to specify the selected program as the main program.



7.4 Executing a program

This section explains how to execute a program existing in the controller.

Step 1 Make sure that the robot has returned to origin position.

In the controller monitor, verify that return-to-origin is complete (the lamp lit). For details on return-to-origin, refer to "4.3 Return-to-origin."



NOTE

All axes connected to the robot must have completed return-to-origin.

Step 2 In the program editor, click [Syntax Check].

In the program editor, click (Syntax Check) to verify that there are no syntax errors. If there are any syntax errors, the syntax check pane shows the name of the program and the line number in which the syntax error was found, and the content of the error. Note the error content and correct the program.

For details on the YAMAHA robot language, refer to the programming manual of the controller you're using. For details on errors, refer to the user's manual of the controller.

Result of syntax check (no error is found)

Syntax Check		
Total Error: 0		
Place of occurrence	Number	Error

Result of syntax check (error is found)

Syntax Check		
(1) Error 1		
Place of occurrence	Number	Error
Program[0:Program]	1	1:1 Syntax error

(Syntax Check)



NOTE

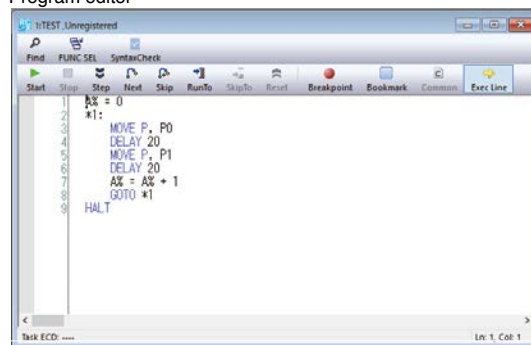
- By selecting an error and double-clicking, you can jump to the location of that error.
- If the syntax check pane is not shown, click (Window)-(Dock All Panes).

Step 3 Execute the program.

In the program editor, click (Start) to execute the program.
To stop the program, click (Stop).



Program editor



■ Other buttons

- [Find]
Opens the [Find and Replace] window. A search string and find or replace the corresponding string can be set.
- [FUNC SEL]
Opens the [Function Selector] window. In the function selector, you can select a function from the list and click [Insert] to insert the selected function at the cursor location of the program editor.
- [Step]
Each time you click this, the command statement at the highlighted line number is executed. If the command statement is a subroutine or a sub-procedure, the first line of that subroutine or sub-procedure is executed.
- [Next]
Each time you click this, the command statement at the highlighted line number is executed. If the command statement is a subroutine or a sub-procedure, the entire subroutine or sub-procedure is executed.
- [Skip]
Each time you click this, you will advance to the next line without executing the highlighted line number.
- [Run To]
Executes up to the cursor location in the program editor.
- [Skip To]
Skips to the cursor position of the program editor without executing the command statements.
- [Reset]
Resets the program. The execution line is returned to the top of the program.
- [Breakpoint]
Sets a break point at the line where the cursor is located in the program editor. If a break point is already specified there, the break point is cleared.
- [Bookmark]
Sets a bookmark at the line where the cursor is located in the program editor. If a bookmark is already specified, the bookmark is cleared.
You can set multiple bookmarks and press the [F2] key to move to the next bookmark or the [Shift+F2] key to move to the previous bookmark.
- [Exec Line]
Enables/disables the line display that is executed.
If the execution line is enabled, the line is highlighted when the program runs.
- [Common]
Switches the display to the common program. When the common program is displayed, pressing [Common] switches the display to the original program.
[Common] is available if [Exec Line] is unavailable and a program named [COMMON] is also saved.
For details about the common program, refer to the programming manual of the controller you're using.



CAUTION

The highlighted execution line display might be later than actual. Consider it an approximate indication of which command statement within the program is currently being executed.

7.5 Editing point data

Use the Point editor to view and edit the controller's point data, point names, and point comments.

Step 1 Open the Point editor.

Use any of the following methods to open the Point editor.

- On the main menu of the target window, click (Coordinate System)-(Point).
- In the Window Tree, double-click (Coordinate System)-(Point).
- In the (Coordinate System Toolbar), click (Point).

Point editor

Name	A1	A2	A3	A4	A5	A6	Arm Type	F1	F2	Comment
P0 SAMPLE1	309.810	355.048	0.000	0.000	-0.007	1.083	None	0	0	Origin position
P1 SAMPLE2	234.694	129.010	1.331	15.797	-0.288	-125.602	None	0	0	Current position
P2	9.800	-55.471	11.099	-0.470	0.000	0.000	Righty	0	0	
P3	327.183	-55.471	61.738	-0.470	0.000	0.000	Righty	0	0	
P4	-0.118	252.897	-0.986	-0.680	0.000	0.000	Lefty	0	0	
P5	-0.120	252.897	30.000	100.650	0.000	0.000	Lefty	0	0	
P6	-101.411	381.339	0.000	250.763	0.000	0.000	Righty	0	0	
P7	-370.552	147.347	0.000	231.504	0.000	0.000	Righty	0	0	
P8	525.391	206.506	-1.000	-52.450	0.000	0.000	Righty	0	0	
P9	-394.513	300.094	500.23	0	0	0				

Step 2 Edit the point data, point names, and point comments.

In the Point editor you can enter point data, point names, and point comments for points between P0 and P29999.

(Arm Type) can be specified only for a SCARA robot.

By clicking (Teach (mm)) or (Teach (pls)) you can also register (teach) the current position of the robot to the selecting point.

Enter the point name in the (Name) column. You can enter up to 16 characters, using single-byte alphanumeric characters and the _ (underscore) character.

Point names must be unique. You can press (Name Check) to check whether any point names are duplicates.

Enter the point comment in the (Comment) column. You can enter up to 16 bytes, and there is no limitation on the types of characters that can be input. For details on points, refer to the manual of the robot controller. When you edit a point, the edited items turn red.

	Name	A1	A2	A3	A4	A5	A6	Arm Type	F1	F2	Comment
P0	SAMPLE1	309.810	355.048	0.000	0.000	-0.007	1.083	None	0	0	Origin position
P1	SAMPLE2	234.694	129.010	1.331	15.797	-0.288	-125.602	None	0	0	Current position
P2	SAMPLE3	10.000	-50.000	10.000	-0.500	1.000	1.000	Lefty	1	1	Target position



NOTE

Execute return-to-origin of the robot before performing teaching input with (Teach (mm)) or (Teach (pls)).

Step 3 Save the edited data.

When you have finished editing the point data, point names, and point comments, click (Edit)-(Save) on the main menu of the target window, or click (Save) in the edit toolbar. When saving is complete, the edited item color returns.

7.6 Editing parameters

Use the Parameter editor to view and edit parameters.

Step 1 Open the Parameter editor.

Use any of the following methods to open the Parameter editor.

- On the main menu of the target window, click (Parameter) and choose the parameter to edit.
- In the Window Tree under (Parameter), double-click the parameter to edit.
- In the (Parameter Toolbar), click the button of the parameter to edit.

Parameter [Controller] editor

Parameter	Value
Incremental Mode control	INVALID
MOVEI/DRIVEI start pos.	KEEP
MOVET start pos.	KEEP
Servo on when power on	UNDO
Sequence flag	0
Safe mode flag	-1
Permission to start program in origin non-completion	INVALID
Permission to reset at program start	INVALID
Permission to reset at controller boot	INVALID
Current program no.	23
Main program no.	23
INPUT/PRINT using channel	PB
Emergency time[ms]	1000
Emergency time2[ms]	300
Debug start mode	LOAD
Break point stop mode	HOLDALL
Shift on area check out	VALID

Step 2 Edit the parameters.

Edit the parameters in the parameter editor.
For items that require to enter a specific setting, a drop-down list appears. For details, refer to the manual of the robot controller.

When editing a parameter, the edited items turn red.

Incremental Mode control	VALID
MOVEI/DRIVEI start pos.	KEEP
MOVET start pos.	KEEP

Step 3 Save the edited parameters.

When you have finished editing the parameters, click (Edit)-(Save) on the main menu of the target window, or click (Save) in the edit toolbar. When saving is complete, the edited item color returns.

7.7 Editing shift data

Use the Shift editor to view and edit the shift data within the controller.

Step 1 Open the Shift editor.

Use any of the following methods to open the Shift editor.

- On the main menu of the target window, click (Coordinate System)-(Shift).
- In the Window Tree, double-click (Coordinate System)-(Shift).
- In the (Coordinate System Toolbar), click (Shift).

Shift editor

	X	Y	Z	R
S0	0.000	0.000	0.000	0.000
SP0	0.000	0.000	0.000	0.000
SM0	0.000	0.000	0.000	0.000
S1	100.000	100.000	50.000	50.000
SP1	75.000	75.000	75.000	75.000
SM1	50.000	50.000	100.000	100.000
S2				
SP2				
SM2				

Step 2 Edit the shift data.

Use the Shift editor to edit the shift data. In addition to the shift coordinates (between S0 and S39), you can also specify a shift coordinate range for each shift coordinate to restrict the robot operating area to a desired range.

The (SPx) value limits the operating range in the plus (+) direction for each axis, and the (SMx) value limits it in the minus (-) direction. For details on shift data, refer to the manual of the robot controller. When editing shift data, the edited items turn red.

	X	Y	Z	R
S0	100.000	100.000	100.000	100.000
SP0	50.000	50.000	50.000	50.000
SM0	0.000	0.000	0.000	0.000

Step 3 Save the edited data.

When you have finished editing the shift data, click (Edit)-(Save) on the main menu of the target window, or click (Save) on the Edit toolbar. When saving is complete, the edited item color returns.

7.8 Editing hand data

Use the Hand editor to view and edit the hand data in the controller.

Step 1 Open the Hand editor.

Use any of the following methods to open the Hand editor.

- On the main menu of the target window, click (Coordinate System)-(Hand).
- In the Window Tree, double-click (Coordinate System)-(Hand).
- In the (Coordinate System Toolbar), click (Hand).

Hand editor

	Robot	Param1	Param2	Param3	R
H0	Robot 1	0.000	0.000	0.000	☒
H1	Robot 1	8.207	0.000	57.766	☒
H2					☐
H3					☐
H4					☐
H5					☐
H6					☐
H7					☐
H8					☐

Step 2 Edit the hand data.

Use the Hand editor to edit the hand data. Hand data can be edited between H0 and H31. For details on hand data, refer to the manual of the robot controller. When editing hand data, the edited items turn red.

	Robot	Param1	Param2	Param3	R
H0	Robot 2	100.000	100.000	100.000	☒
H1	Robot 1	8.207	0.000	57.766	☒
H2					☐

Step 3 Save the edited data.

When you have finished editing the hand data, click (Edit)-(Save) on the main menu of the target window, or click (Save) on the Edit toolbar. When saving is complete, the edited item color returns.

7.9 Editing work data

Use the Work editor to view and edit the area check output data.

Step 1 Open the Work editor.

Use any of the following methods to open the Work editor.

- On the main menu of the target window, click (Coordinate System)-(Work).
- In the Window Tree, double-click (Coordinate System)-(Work).
- In the (Coordinate System Toolbar), click (Work).

Work editor

	X axis offset	Y axis offset	Z axis offset	R axis offset
W0	10.000	20.000	30.000	40.000
W1	100.000	150.000	180.000	200.000
W2	35.000	55.000	75.000	95.000
W3				
W4				
W5				
W6				
W7				
W8				

Step 2 Edit the work data.

Use the Work editor to edit the work data. Set the (X, Y, Z, and R-axis (rotation)) offset from the robot tip or hand tip. When editing work data, the edited items turn red.

	X axis offset	Y axis offset	Z axis offset	R axis offset
W0	20.000	30.000	40.000	50.000
W1	100.000	150.000	180.000	200.000
W2	35.000	55.000	75.000	95.000

Step 3 Save the edited data.

When you have finished editing the work data, click (Edit)-(Save) on the main menu of the target window, or click (Save) on the Edit toolbar. When saving is complete, the edited item color returns.

7.10 Editing pallet data

Use the Pallet editor to view and edit the pallet data in the controller.

Step 1 Open the Pallet editor.

Use any of the following methods to open the Pallet editor.

- On the main menu of the target window, click (Coordinate System)-(Pallet).
- In the Window Tree, double-click (Coordinate System)-(Pallet).
- In the (Coordinate System Toolbar), click (Pallet).

Pallet editor

Edit Point	PLN	NX	NY	NZ	PLP
PL0					
PL1	XY	5	5	1	0
PL2					
PL3					
PL4					
PL5					
PL6					
PL7					

Step 2 Edit the pallet data.

Use the Pallet editor to edit the pallet data. Pallet data can be edited between PL0 and PL39. For details on pallet data, refer to the manual of the robot controller.

	PLN	NX	NY	NZ	PLP
PL0	YZ	1	1	5	5
PL1	XY	5	5	1	0
PL2					

When you click (Edit Point), the Point editor opens with the point number specified by PLP selected. If the Point editor is already open, you will move to the point with the point number specified by PLP. When editing pallet data, the edited items turn red.

Step 3 Save the edited data.

When you have finished editing the pallet data, click (Edit)-(Save) on the main menu of the target window, or click (Save) on the Edit toolbar. When saving is complete, the edited item color returns.

7.11 Editing area check output data

Use the Area Check Output editor to view and edit the area check output data.

Step 1 Open the Area Check Output editor.

Use any of the following methods to open the Area Check Output editor.

- On the main menu of the target window, click (Coordinate System)-(Area Check Output).
- In the Window Tree, double-click (Coordinate System)-(Area Check Output).
- In the (Coordinate System Toolbar), click (Area Check Output).

Area Check Output editor

	Robot	Point 1	Point 2	Port Type	Port No.	Logic
AC0	Robot 1	10	11	SO	20	ON
AC1	Robot 1	12	13	SO	21	ON
AC2	Robot 1	14	15	SO	22	ON
AC3	Robot 1	16	17	SO	23	ON
AC4	Robot 1	18	19	SO	24	ON
AC5	Robot 1	20	21	SO	25	ON
AC6	Robot 1	22	23	SO	26	ON
AC7	Robot 1	24	25	SO	27	ON
AC8	Robot 1	26	27	SO	30	ON
AC9	Robot 1	28	29	SO	31	ON
AC10						

Step 2 Edit the area check output data.

Use the Area Check Output editor to edit the area check output data which can be edited between AC0 and AC31.

	Robot	Point 1	Point 2	Port Type	Port No.	Logic
AC0	Robot 2	20	12	MO	30	Off
AC1	Robot 1	12	13	SO	21	ON
AC2	Robot 1	14	15	SO	22	ON

For details on area check output data, refer to the manual of the robot controller.

When editing area check output data, the edited items turn red.

Step 3 Save the edited data.

When you have finished editing the area check output data, click (Edit)-(Save) on the main menu of the target window, or click (Save) on the Edit toolbar. When saving is complete, the edited item color returns.

7.12 Editing general Ethernet port data

Use the General Ethernet Port editor to view and edit general Ethernet port data.

TIP

For details on general Ethernet ports, refer to the manual of the robot controller.

Step 1 Open the General Ethernet Port editor.

Use any of the following methods to open the General Ethernet Port editor.

- On the main menu of the target window, click (Parameter)-(General Ethernet Port).
- In the Window Tree, double-click (Parameter)-(General Ethernet Port).
- In the (Parameter Toolbar), click (General Ethernet Port).

General Ethernet Port editor

	Mode	IP Address	Port	End of Line
GP0	Server	133.215.205.26	30000	CRLF
GP1				
GP2				
GP3	Client	133.215.205.26	30000	CRLF
GP4				
GP5				
GP6				
GP7				

Step 2 Edit the general Ethernet port data.

Use the General Ethernet Port editor to edit the general Ethernet port data which can be edited between GP0 and GP7.

When editing general Ethernet port data, the edited items turn red.

	Mode	IP Address	Port	End of Line
GP0	Client	133.215.205.26	10000	CR
GP1				
GP2				
GP3	Client	133.215.205.26	30000	CRLF

Step 3 Save the edited data.

When you have finished editing the general Ethernet port data, click (Edit)-(Save) on the main menu of the target window, or click (Save) on the Edit toolbar. When saving is complete, the item color returns.

7.13 Editing camera calibration data

Use the Camera Calibration editor to view and edit Camera Calibration data.

Step 1 Open the Camera Calibration editor.

Use any of the following methods to open the Camera Calibration editor.

- On the main menu of the target window, click (Coordinate System)-(Camera Calibration).
- In the Window Tree, double-click (Coordinate System)-(Camera Calibration).
- In the (Coordinate System Toolbar), click (Camera Calibration).

Camera Calibration editor

Camera Calibration									
Worksheet									
Robot	Installation method	Camera Direction	Scale Ratio	Camera Angle	Parameter1	Parameter2	Number of Pixels (H)	Number of Pixels (V)	
CL8	Robot 1	Fixed	Up Direction	0.0004	2.289	249.677	282.922	1624	1236
CL1	Robot 1	Fixed	Up Direction	0.0489	2.289	249.677	282.922	1624	1236
CL2									
CL3									
CL4									
CL5									
CL6									
CL7									

Step 2 Edit the camera calibration data.

Use the Camera Calibration editor to edit the camera calibration data. For details, refer to the iVY2 System manual. When editing camera calibration data, the edited items turn red.

	Robot	Installation method	Camera Direction	Scale Ratio	Camera Angle	Parameter1	Parameter2	Number of Pixels (H)	Number of Pixels (V)
CL8	Robot 1	Fixed	Up Direction	0.0004	2.230	249.677	282.922	1624	1236
CL1	Robot 1	Fixed	Up Direction	0.0489	2.239	249.677	282.922	1624	1236
CL2	Robot 2	2nd Arm	Down Direction	0.5000	2.200	250.000	288.000	1624	1236

Step 3 Save the edited data.

When you have finished editing the camera calibration data, click (Edit)-(Save) on the main menu of the target window, or click (Save) on the Edit toolbar. When saving is complete, the item color returns.

7.14 Editing conveyor calibration data

Use the Conveyor Calibration editor to view and edit Conveyor Calibration data.

Step 1 Open the Conveyor Calibration editor.

Use any of the following methods to open the Conveyor Calibration editor.

- On the main menu of the target window, click (Coordinate System)-(Conveyor Calibration).
- In the Window Tree, double-click (Coordinate System)-(Conveyor Calibration).
- In the (Coordinate System Toolbar), click (Conveyor Calibration).

Conveyor Calibration editor

	Robot1	Robot2	Robot3	Robot4
X	10.000	30.000		
Y	20.000	10.000		
Z	30.000	20.000		
R	5.000	10.000		
Upstream Pos.	15.000	5.000		
Downstream Pos.	25.000	10.000		

Step 2 Edit the conveyor calibration data.

Use the Conveyor Calibration editor to edit the conveyor calibration data. For details, refer to the Tracking System manual.

When editing conveyor calibration data, the edited items turn red.

	Robot1	Robot2	Robot3	Robot4
X	20.000	30.000		
Y	30.000	10.000		
Z	10.000	20.000		

Step 3 Save the edited data.

When you have finished editing the conveyor calibration data, click (Edit)-(Save) on the main menu of the target window, or click (Save) on the Edit toolbar. When saving is complete, the item color returns.

Revision record

Issue date	Description	
Apr. 2020	Ver. 1.00	First edition
June. 2020	Ver. 1.01	Changed descriptions of 'USB key' in "1. About RCX-Studio 2020".
June. 2022	Ver. 1.02	Added PCVision function.
Feb. 2023	Ver. 1.03	Changed descriptions of 'USB key' in "1. About RCX-Studio 2020". Added "4.8 Importing/Exporting target".
Sep. 2023	Ver. 1.04	Updated the supported Windows OS version.
Jan. 2025	Ver. 1.05	Added the controller RCX341.

User's Manual

Support Software

RCX-Studio 2020

January 2025

Ver. 1.05

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